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United States Patent	12403611
Kind Code	B2
Date of Patent	September 02, 2025
Inventor(s)	McCall; David et al.

Head and neck assembly for a humanoid robot

Abstract

A humanoid robot includes an upper region includes a head and neck assembly having a neck portion and a head portion coupled to the neck portion. Said head portion includes: a rear shell, and a frontal shell coupled to the rear shell to define a head volume between the frontal shell and the rear shell, and an electronics assembly with a display located in the head volume between the frontal shell and the rear shell. The frontal shell of the head housing assembly includes: a first arc length at a first location below the display, a second arc length at a second location aligned with a portion of the display, and wherein said second arc length is greater than the first arc length, and a third arc length at a third location above the display, and wherein said third arc length is greater than both the first arc length and the second arc length.

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Family ID: **95068587**

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Appl. No.: **18/919274**

Filed: **October 17, 2024**

Prior Publication Data

Document Identifier	Publication Date
US 20250100156 A1	Mar. 27, 2025

Related U.S. Application Data

continuation-in-part parent-doc US 18914800 20241014 PENDING child-doc US 18919274
continuation-in-part parent-doc US 18904332 20241002 PENDING child-doc US 18914800
continuation-in-part parent-doc US 29935680 20240403 PENDING child-doc US 18914800

continuation-in-part parent-doc US 29928748 20240215 PENDING child-doc US 18904332
continuation-in-part parent-doc US 29889764 20230417 PENDING child-doc US 29928748
us-provisional-application US 63707897 20241016
us-provisional-application US 63707949 20241016
us-provisional-application US 63708003 20241016
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us-provisional-application US 63626028 20240227
us-provisional-application US 63626035 20240227
us-provisional-application US 63626030 20240221

Publication Classification

Int. Cl.: **B25J11/00** (20060101); **H04N23/90** (20230101)

U.S. Cl.:

CPC **B25J11/0015** (20130101); H04N23/90 (20230101)

Field of Classification Search

USPC: None

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Primary Examiner: Braniff; Christopher

Background/Summary

PRIORITY CLAIM AND CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is: (i) a continuation in part of U.S. patent application Ser. No. 18/914,800, filed Oct. 14, 2024, which is a continuation in part of U.S. patent application Ser. No. 18/904,332, filed Oct. 2, 2024, (ii) a continuation in part of U.S. Design patent application Ser. No. 29/935,680, filed Apr.

3, 2024, which is a continuation in part of U.S. Design patent application Ser. No. 29/928,748, filed Feb. 15, 2024, which is a continuation in part of U.S. Design patent application Ser. No. 29/889,764, filed Apr. 17, 2023, and (iii) claims priority under 35 U.S.C. § 119 (e) to U.S. Provisional Patent Application No. 63/626,035, filed Feb. 27, 2024, U.S. Provisional Patent Application No. 63/564,741, filed Mar. 13, 2024, U.S. Provisional Patent Application No. 63/626,034, filed Mar. 13, 2024, and U.S. Provisional Patent Application No. 63/626,037, filed May 28, 2024, U.S. Provisional Patent Application No. 63/626,030, filed Feb. 21, 2024, U.S. Provisional Patent Application No. 63/566,595, filed Mar. 18, 2024, U.S. Provisional Patent Application No. 63/626,028, filed Feb. 27, 2024, U.S. Provisional Patent Application No. 63/573,528, filed Apr. 3, 2024, U.S. Provisional Patent Application No. 63/561,316, filed Mar. 5, 2024, U.S. Provisional Patent Application No. 63/634,697, filed Apr. 16, 2024, U.S. Provisional Patent Application No. 63/573,226, filed Apr. 2, 2024, U.S. Provisional Patent Application No. 63/707,949, filed Oct. 16, 2024, U.S. Provisional Patent Application No. 63/707,897, filed Oct. 16, 2024, U.S. Provisional Patent Application No. 63/707,547, filed Oct. 16, 2024, U.S. Provisional Patent Application No. 63/708,003, filed Oct. 16, 2024, each of which is expressly incorporated by reference herein in its entirety. (2) Reference is hereby made to U.S. Provisional Patent Application No. 63/557,874, filed Feb. 26, 2024, U.S. Provisional Patent Application No. 63/626,040, filed Jan. 28, 2024, U.S. Provisional Patent Application No. 63/626,105, filed Jan. 29, 2024, U.S. Provisional Patent Application No. 63/625,362, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,370, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,381, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,384, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,389, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,405, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,423, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/625,431, filed Jan. 26, 2024, U.S. Provisional Patent Application No. 63/685,856, filed Aug. 22, 2024, U.S. Provisional Patent Application No. 63/696,507, filed Sep. 19, 2024, U.S. Provisional Patent Application No. 63/696,533, filed Sep. 19, 2024, and U.S. Provisional Patent Application No. 63/706,768, filed Oct. 14, 2024, each of which is expressly incorporated by reference herein in its entirety.

TECHNICAL FIELD

(1) This disclosure relates to a head of a robot, specifically a head of a humanoid robot. The head of the humanoid robot includes a plurality of components configured to provide the robot with the ability to communicate with nearby humans using a display that is protected by a frontal shell.

BACKGROUND

(2) The current labor market within the United States is confronting an unprecedented labor shortage, characterized by over 10 million unfilled positions. A significant proportion of these vacancies pertain to occupations that are deemed unsafe, undesirable, or involve hazardous working conditions. This persistent and escalating shortage of available labor has created an urgent imperative for the development and deployment of advanced robotic systems capable of performing tasks that are unattractive or pose risks to human workers. To effectively address this widening gap in the workforce, it has become critical to design and engineer robots that can operate with high efficiency and reliability within human-centric environments. These environments often demand capabilities such as physical dexterity, sustained endurance, precise manipulation, and the ability to navigate complex spaces designed for humans.

(3) Advanced general-purpose humanoid robots have emerged as a promising solution to meet these challenges. These robots are meticulously engineered to replicate the human form and emulate human functionality, typically featuring bipedal locomotion with two legs, bilateral manipulation abilities with two arms, and a display to facilitate interaction with human users. The anthropomorphic design enables these robots to seamlessly integrate into environments originally designed for humans, thereby minimizing the need for extensive modifications to existing

infrastructures. As these robots endeavor to mimic the human body, it becomes essential to equip them with a head design that not only meets functional requirements but also enhances aesthetic appeal and durability. The head is a critical component for human-robot interaction, serving as the primary interface through which the robot communicates and engages with nearby humans. A well-designed head can significantly improve the robot's ability to convey information, express intentions, and respond to human cues, thereby fostering a more intuitive and natural interaction experience.

(4) To meet these requirements, the present disclosure introduces an innovative head design that incorporates a versatile display system. This display is capable of adapting its visual output to suit a wide range of operational tasks by rendering icons, graphics, expressive animations, and informative text. The adaptability of the display allows the robot to present contextually relevant information and provide visual feedback, all of which enhance the robot's ability to interact effectively with human users. By making the robot's appearance more relatable and intuitive, the display fosters improved engagement and facilitates smoother human-robot collaboration.

(5) Considering the sensitive and fragile nature of display technologies, and acknowledging the often challenging and harsh environments in which humanoid robots are deployed, it is advantageous to position the display behind a protective shield. This strategic placement serves multiple purposes. Firstly, the shield safeguards the display from potential contaminants such as dust, moisture, chemicals, and particulate matter that could adversely affect its performance and longevity. Secondly, the shield provides protection against physical impacts, vibrations, and mechanical stresses that may occur during operation, especially in industrial or outdoor settings. By mitigating the risks of damage to the display, the shield contributes to the overall robustness and reliability of the robot. Moreover, the integration of the display behind a shield contributes to a sleek and futuristic aesthetic, enhancing the robot's visual appeal.

(6) In summary, the disclosed head design addresses the critical need for a durable, adaptable, and aesthetically pleasing interface for a general-purpose humanoid robot. By combining a versatile display with the frontal shell, the design ensures that the robot can effectively communicate and interact with humans while withstanding the rigors of diverse operational environments. This innovation not only enhances the functionality and user experience but also extends the operational lifespan of the robot, thereby providing a more sustainable and cost-effective solution for addressing the current labor market challenges.

SUMMARY

(7) A need exists for a humanoid robot with an upper region including: (i) a torso, (ii) a pair of arm assemblies coupled to the torso, and (iii) a head and neck assembly coupled to the torso and having a neck portion and a head portion coupled to the neck portion. Said head portion includes: a frontal shell having a rear edge, a rear shell having a frontal edge, and an illumination assembly. The illumination assembly is configured to illuminate a region that: (i) extends between a rear edge of the frontal shell and an extent of the frontal edge of the rear shell, (ii) is positioned adjacent to the extent of the rear edge of the frontal shell, and (iii) is positioned adjacent to the extent of the frontal edge of the rear shell. The humanoid robot also includes: (i) a central region coupled to the upper region, and (ii) a lower region coupled to the central region and spaced apart from the upper region, the lower region including a pair of legs.

(8) There is also a need for a humanoid robot with an upper region including: (i) a torso, (ii) a pair of arm assemblies coupled to the torso, and (iii) a head and neck assembly coupled to the torso and having a neck portion and a head portion coupled to the neck portion. Said head portion includes a frontal shell having a curvilinear periphery, and an outer surface having a nasal region and an orbital region that is not recessed in comparison to said nasal region. The head portion also includes an illumination assembly configured to emit light in a location that is adjacent to the periphery of the frontal shell. Finally, the humanoid robot also includes: (i) a central region coupled

to the upper region, and (ii) a lower region coupled to the central region and spaced apart from the upper region, the lower region including a pair of legs.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The drawing figures depict one or more implementations in accord with the present teachings, by way of example only, not by way of limitation. In the figures, like reference numerals refer to the same or similar elements shown across various other figures.

(2) FIG. **1** is a front perspective view of a humanoid robot in an upright, standing position P**1** and including: (i) an upper region having the following parts: (a) a head and neck assembly, (b) a torso, (c) left and right shoulders, (d) and left and right arm assemblies each including: (e) a humerus, (f) a forearm, (g) a wrist, and (h) a hand; (ii) a lower region having left and right leg assemblies each including: (a) a thigh, (b) a knee, (c) a shin, (d) an ankle, and (e) a foot; and (iii) a central region connecting the upper portion and the lower portion to one another and configured to allow movement of the upper and lower regions relative to one another;

(3) FIG. **2** is a rear perspective view of the humanoid robot of FIG. **1** in the upright, standing position P**1**;

(4) FIG. **3** is a perspective view of the head and neck assembly of the humanoid robot of FIG. **1** showing the head and neck assembly including: (i) a head portion having: (a) a head housing assembly with a frontal shell and a rear shell, (b) a head electronics assembly with an illumination assembly having at least one light emitter configured to emit light between an extent of the frontal shell and rear shell, and (ii) a neck portion having: (a) a neck housing assembly with a cover, and (b) a neck electronics assembly;

(5) FIG. **4** is a front view of the head and neck assembly of FIG. **3** showing: (i) the frontal shell coupled to the rear shell, and (ii) a plurality of light emitters of the illumination assembly located at a junction between the frontal shell and the rear shell;

(6) FIG. **5**A is a side view of the head and neck assembly of FIG. **3**;

(7) FIG. **5**B is a side view of the head and neck assembly of FIG. **3** showing various anatomical regions of the head;

(8) FIG. **6** is a rear view of the head and neck assembly of FIG. **3**;

(9) FIG. **7** is a top view of the head and neck assembly of FIG. **3**;

(10) FIG. **8** is a partial perspective view of the head and neck assembly of FIG. **3** showing a first set of information displayed on and/or through the frontal shell;

(11) FIG. **9**A is a front view of the head and neck assembly of FIG. **3** showing a second set of information displayed on an/or through the frontal shell;

(12) FIG. **9**B is a front view of the head of the robot showing an icon indicating a battery status on an/or through the frontal shell;

(13) FIG. **9**C is a front view of the head of the robot showing another icon indicating a low battery status on an/or through the frontal shell;

(14) FIG. **9**D is a front view of the head of the robot showing another icon indicating an alert or system failure event on an/or through the frontal shell;

(15) FIG. **9**E is a front view of the head of the robot showing an icon indicating a particular mode of the robot on an/or through the frontal shell;

(16) FIG. **9**F is a front view of the head of the robot showing an icon indicating a particular task being performed by the robot on an/or through the frontal shell;

(17) FIG. **9**G is a front view of the head of the robot showing an additional robot status icon on an/or through the frontal shell;

(18) FIG. **10** is a perspective view of a second embodiment of a humanoid robot in an upright

standing position P**1** with outer covers of the robot removed to expose various actuators and components included in the robot;
(19) FIG. **11** is a front view of the humanoid robot of FIG. **10**;
(20) FIG. **12** is a perspective view of the head and neck assembly of FIG. **10**, wherein the head electronics assembly further includes a display that is visible through the frontal shell;
(21) FIG. **13** is a bottom perspective view of the head and neck assembly of FIG. **12**;
(22) FIG. **14** is a front view of the head and neck assembly of FIG. **12** showing the head and neck assembly including: (i) the display, and (ii) a plurality of light emitters of the illumination assembly that are positioned to emit light on the lateral sides of the head and adjacent to a rear edge of the frontal shell;
(23) FIG. **15**A is a side view of the head and neck assembly of FIG. **12**;
(24) FIG. **15**B is a side view of the head and neck assembly of FIG. **12** showing various anatomical regions of the head;
(25) FIG. **16** is an exploded assembly view of the head and neck assembly of FIG. **12** wherein: (i) the head housing assembly includes an electronics support and an intermediate cover, (ii) the display is coupled to the electronic support, and (iii) the light emitters of the illumination assembly are coupled to the intermediate cover;
(26) FIG. **17** is a perspective view of the display of FIG. **12**;
(27) FIG. **18** is a front view of the display of FIG. **12**;
(28) FIG. **19** is a top view of the display of FIG. **12**;
(29) FIG. **20** is a side view of the display of FIG. **12**;
(30) FIG. **21** is a top view of the display of FIG. **12**, wherein a circle is disposed on the inside of the display in order to show its curvilinear configuration;
(31) FIG. **22** is a side view of the display showing a plurality of zones in a horizontal arrangement relative to one another;
(32) FIG. **23** is a front view of the display of FIG. **22**;
(33) FIG. **24** is a side view of the display showing a plurality of zones in a vertical arrangement relative to one another;
(34) FIG. **25** is a front view of the display of FIG. **24**;
(35) FIG. **26** is a perspective view of the head and neck assembly of FIG. **12**, wherein the head electronics assembly includes a sensor assembly: (i) coupled to the electronics support, (ii) positioned above the display, and (iii) within the interior space formed between the intermediate cover and the frontal shell;
(36) FIG. **27** is a front view of the head and neck assembly of FIG. **26**;
(37) FIG. **28** is a rear perspective view of a second embodiment of a head and neck assembly of the humanoid robot of FIG. **1**;
(38) FIG. **29** is a front perspective view of the head and neck assembly of FIG. **28**, wherein the head assembly further includes an upper shell with an upper recessed sensor zone;
(39) FIG. **30** is a front view of the head and neck assembly of FIG. **28**;
(40) FIG. **31**A is a side view of the head and neck assembly of FIG. **28**;
(41) FIG. **31**B is a side view of the head and neck assembly of FIG. **28** showing various anatomical regions of the head;
(42) FIG. **32** is a perspective view of a third embodiment of head and neck assembly of the humanoid robot, wherein the head and neck assembly includes a deformable mesh covering internal structures and components of the neck assembly;
(43) FIG. **33** is a lower perspective view of the third embodiment of the head and neck assembly of FIG. **32**, wherein the head and neck assembly includes: (i) a head portion having: (a) a head housing assembly with a frontal shell, a rear shell, and an upper shell with an upper recessed sensor zone, (b) a head electronics assembly with an illumination assembly having at least one light emitter configured to emit light between an extent of the frontal shell and rear shell, and (ii) a neck

portion having: (a) a neck housing assembly with a deformable cover and a lower recessed sensor zone, and (b) a neck electronics assembly;
(44) FIG. **34** is a bottom perspective view of the head and neck assembly of FIG. **32**;
(45) FIG. **35** is a side perspective view of the head and neck assembly of FIG. **32**;
(46) FIG. **36** is a front view of the head and neck assembly of FIG. **32**;
(47) FIG. **37** is a side view of the head and neck assembly of FIG. **32**;
(48) FIG. **38** is a rear view of the head and neck assembly of FIG. **32**;
(49) FIG. **39** is a top view of the head and neck assembly of FIG. **32**;
(50) FIG. **40** is a bottom view of the head and neck assembly of FIG. **32**;
(51) FIG. **41** is a top perspective view of the head and neck assembly of FIG. **32** wherein: (i) the frontal shell is shown as transparent to reveal the display located in an interior space defined between the frontal shell and rear shell, and (ii) an extent of the upper sensor zone is shown as transparent to reveal a pair of cameras located in the interior space and positioned above the display;
(52) FIG. **42** is a side view of the head and neck assembly of FIG. **41**;
(53) FIG. **43** is a perspective view of the frontal shell of the head and neck assembly of FIG. **41**, wherein the frontal shell includes: (i) a main body, (ii) a plurality of wings that extend upward from the main body, (iii) an outer periphery with a rear edge having a plurality of recesses formed therein;
(54) FIG. **44** is a side view of the frontal shell of the head and neck assembly of FIG. **41** showing various dimensions of the frontal shell;
(55) FIG. **45** is a front view of the frontal shell of the head and neck assembly of FIG. **41**;
(56) FIG. **46** is a cross-sectional view of the frontal shell of FIG. **41** taken along line **46-46** in FIG. **45** and showing three different radii of curvatures;
(57) FIG. **47** is a cross-sectional view of the frontal shell of FIG. **41** taken along line **47-47** in FIG. **45** and showing three different radii of curvatures;
(58) FIG. **48** is a cross-sectional view of the frontal shell of FIG. **41** taken along line **48-48** in FIG. **45** and showing three different radii of curvatures;
(59) FIG. **49** is a cross-sectional view of the frontal shell of FIG. **41** taken along line **49-49** in FIG. **45** and showing three different radii of curvatures;
(60) FIG. **50** is a front view of the frontal shell with a plurality of zones in a horizontal arrangement relative to one another;
(61) FIG. **51** is a side view of the frontal shell of FIG. **50**;
(62) FIG. **52** is a front view of the frontal shell with a plurality of zones in a vertical arrangement relative to one another;
(63) FIG. **53** is a side view of the frontal shell of FIG. **52**;
(64) FIG. **54** is a perspective view of the third embodiment of the head and neck assembly wherein the frontal shell is omitted from the housing assembly to further show the display positioned within said head assembly;
(65) FIG. **55** is a front view of the head and neck assembly of FIG. **54**;
(66) FIG. **56** is a bottom perspective view of the head and neck assembly of FIG. **54**;
(67) FIG. **57**A is a side view of the head and neck assembly of FIG. **54**;
(68) FIG. **57**B is a side view of the head and neck assembly of FIG. **57**A showing various anatomical regions of the head;
(69) FIG. **58** is a perspective view of a fourth embodiment of a head and neck assembly, wherein the frontal shell is omitted from the housing assembly, the electronics frame further includes a plurality of support ribs, and the electronics assembly further includes a plurality of side displays coupled to the electronics frame;
(70) FIG. **59** is a bottom perspective view of the head and neck assembly of FIG. **58**;
(71) FIG. **60** is a perspective view of a fifth embodiment of a head and neck assembly, wherein the

illumination assembly includes a plurality of light emitters that are positioned adjacent to a majority of the rear edge of said frontal shell;
(72) FIG. **61** is a perspective view of a sixth embodiment of a head and neck assembly, wherein the upper recessed sensor zone of the fifth embodiment has been omitted;
(73) FIG. **62** is a side view of a seventh embodiment of a head and neck assembly wherein the rear shell has been omitted and the frontal shell encases a majority of the head;
(74) FIG. **63**A is an enlarged view of a portion of a torso region of the humanoid robot of FIG. **1**, wherein said torso includes a plurality of light emitters;
(75) FIG. **63**B is an enlarged view of a portion of a right thigh region of the humanoid robot of FIG. **1**, wherein said thigh region includes at least one light emitter;
(76) FIG. **63**C is an enlarged view of a portion of a left thigh region of the humanoid robot of FIG. **1**, wherein said thigh region includes at least one light emitter;
(77) FIG. **64** is an enlarged view of a portion of the head and neck region of the humanoid robot of FIG. **6**, wherein said neck includes at least one light emitter;
(78) FIG. **65**A is an enlarged view of a portion of a shoulder region of the humanoid robot of FIG. **32**, wherein said shoulder region includes at least one light emitter;
(79) FIG. **65**B is an enlarged view of a portion of a wrist region of the humanoid robot of FIG. **32**, wherein said wrist region includes at least one light emitter; and
(80) FIG. **65**C is an enlarged view of a portion of a knee region of the humanoid robot of FIG. **32**, wherein said knee region includes at least one light emitter.

DETAILED DESCRIPTION

(81) In the following detailed description, numerous specific details are set forth by way of examples in order to provide a thorough understanding of the relevant teachings. However, it should be apparent to those skilled in the art that the present teachings may be practiced without such details. In other instances, well-known methods, procedures, components, and/or circuitry have been described at a relatively high-level, without detail, in order to avoid unnecessarily obscuring aspects of the present disclosure.
(82) While this disclosure includes several embodiments in many different forms, there is shown in the drawings and will herein be described in detail embodiments with the understanding that the present disclosure is to be considered as an exemplification of the principles of the disclosed methods and systems, and is not intended to limit the broad aspects of the disclosed concepts to the embodiments illustrated. As will be realized, the disclosed methods and systems are capable of other and different configurations and several details are capable of being modified all without departing from the scope of the disclosed methods and systems. For example, one or more of the following embodiments, in part or whole, may be combined consistent with the disclosed methods and systems. As such, one or more steps from the flow charts or components in the Figures may be selectively omitted and/or combined consistent with the disclosed methods and systems. Additionally, one or more steps from the flow charts or the method of assembling the shoulder and upper arm may be performed in a different order. Accordingly, the drawings, flow charts and detailed description are to be regarded as illustrative in nature, not restrictive or limiting.

A. Introduction

(83) Unlike conventional robots and as described in greater detail below, the disclosed humanoid robot **100** includes an upper region **200** having a head and neck assembly **202**. The head and neck assembly **202** is coupled to a torso **204** and has an overall shape that generally resembles a human head. As such, the head and neck assembly **202** includes a head portion **202***a*, **2202***a*, **3202***a*, **4202***a*, **5202***a*, **6202***a*, **7202***a* that does not include large flat surfaces (e.g., opposed sides of a head, or is not in the shape of: (a) a cube, (b) a hexagonal prism, or (c) a pentagonal prism). Instead, almost all the surfaces of the head portion **202***a* are curvilinear or have substantial curvilinear aspects or segments. However, as shown in the Figures, some embodiments of the head portion **202***a* do include a recess with a small flat sensor cover or lens, which is recessed in a top portion of

the head portion **202***a* and is designed to decrease sensor signal distortion that may be caused if the sensor signals are required to travel through a curvilinear shell, cover, shield, or lens. Additionally, while the overall head shape is designed to be human-like, the disclosed head portion **202***a* lacks pronounced human facial structures (cheeks, eye peripheral protrusions, a mouth, or other moving structures).

(84) The frontal region of the head portion **202***a* is covered by a large freeform frontal shell, frontal head shell, or frontal shield **228**, **2228**, **3228**, **4228**, **5228**, **6228**, **7228** wherein the curvature of the frontal shell **228** varies both horizontally and laterally across the head portion **202***a*. The freeform nature of the frontal shell **228** causes it to be a separate and distinct component from the display(s) **300**, **2300**, **3300**, **4300**, **5300**, **6300**, **7300** that is positioned behind the frontal shell **228**. This positional relationship allows the frontal shell **228** to protect the display **300** and other electronics contained within the head portion **202***a* from potential damage, a feature which provides a substantial benefit over conventional robot heads that lack this protective element. For example, certain tasks, that the humanoid robot **100** may perform on a factory floor such as moving and cutting sheet metal, may damage or break a display **300** that is not adequately protected behind such a frontal shell **228**. As shown in the Figures, the frontal shell **228** does not extend over the entirety of an upper head shell, nor behind an ear region, nor does it extend into the rear region of the head portion **202***a*. However, the frontal shell **228** extends to the chin region of the head portion **202***a* and, in some embodiments, includes a substantial opening or recess formed along the upper extent of the shield **228**. The opening or recess, which is formed along the upper edge of the frontal shell **228**, allows for the inclusion of a small flat sensor cover or a secondary lens. Due to the formation of this opening or recess, the frontal shell **228** includes two wing-shaped structures that extend upwards from a main body of the shell and effectively surround lateral extents of the sensors that are positioned behind the aforementioned small flat sensor cover.

(85) Unlike conventional robot heads, the disclosed head portion **202***a* includes a display **300**, **2300**, **3300**, **4300**, **5300**, **6300**, **7300** that is preferably curved in a single direction, or at least one direction, and is positioned on an angle relative to the coronal plane and a horizontal reference plane. The curved nature of the display **300** allows for the inclusion of a larger display with a larger surface area within the head portion **202***a*; this, in turn, increases the amount of information that can be displayed on the display **300**. The larger display **300** provides a benefit over conventional robot heads that lack this feature because those conventional robots must either forgo displaying as much information (while not altering the size of the information) or undesirably increase the size of their head (which consequently causes a number of other issues, including increased material costs and assembly costs). Additionally, being able to display more information on the disclosed display **300** is beneficial because the disclosed humanoid robot **100** does not include any other internal displays. Further, including only a single display **300** within the humanoid robot **100** is beneficial because it: (i) reduces the space needed for displays, (ii) reduces the battery usage attributable to displays, and (iii) at least reduces, and typically eliminates, the inclusion of fragile components within the robot. The display may be configured to display robot status, sensor data, and/or other relevant information to nearby human beings. However, the display is not configured to display human-like facial features (eyes, nose, mouth, etc.) or expressions, but instead is designed to use generic blocks or shapes.

(86) Unlike conventional robot heads, in some embodiments, the disclosed head portion **202***a* may include two separate sensor assemblies **301***a*, **2301***a*, **2301***c*, **3301***a*, **3301***b*, **4301***a*, **5301***a*. The first camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **4302**, **5302** may be positioned within the upper shell or the robot's forehead region, while the second camera **3305** may be positioned within the neck portion **202***b* or the robot's chin region. The strategic position of the first camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **4302**, **5302** offers two advantages: (i) it enables a larger display **300** to be utilized within the head portion **202***a*, and (ii) it allows the humanoid robot **100** to see into a bin that is placed on a high shelf or rack. Including the second camera **3305** enables the humanoid

robot **100** to look downward (e.g., to see what it is carrying or to look into a storage bin) without needing to use the first camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **4302**, **5302**. These features are significantly beneficial over conventional robots that lack a second camera **3305**, because such conventional robots must bend and articulate their neck to a greater degree to obtain the data that would otherwise be captured by the second camera **3305**. Also, neither camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **4302**, **5302** in the disclosed head portion **202***a* is positioned where a human's eyes would typically be located, nor above the crown of the head, nor on either side of the robot's head **202***a*. It should be understood that various configurations are possible: (i) both the first and second camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **3305**, **4302**, **5302** may be omitted, (ii) the first or upper camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **4302**, **5302** may be omitted, while the second or lower camera **3305** is retained, or (iii) the second or lower camera **3305** may be omitted, while the first or upper camera(s) **302**, **303**, **304**, **2302**, **3303**, **3304**, **4302**, **5302** is retained.
(87) The electronics assembly of the disclosed head portion **202***a* may include an illumination assembly **263** having at least one light emitter, and preferably a plurality of light emitting assemblies **264***a*-**264***d*, **2264***a*-**2264***d*, **3264***a*-**3264***d*, **4264***a*-**4264***d*, **5264**, **6264**, **7264** are positioned adjacent to a rear edge of the frontal shell **228**. These light emitters **264**.**2***a*-**264**.**2***d* enable the humanoid robot **100** to communicate with humans without using the display **300** that is disposed behind the frontal shell **228**, wherein said light emitters act as, or are configured to act as, indicator lights. Typically, the light emitters (and in this configuration, these indicator lights) can communicate information about the humanoid robot **100** to nearby humans by: (i) emitting light having different wavelengths, wherein said emitted light may be perceived by a nearby human as having different colors of light, and/or (ii) utilizing various illumination sequences, durations, and/or brightness levels. For example, the indicator lights may be used to communicate the working state (e.g., yellow light—600 nm), idle state (e.g., green light 550 nm), charging state (e.g., blinking light or white light), error state (e.g., red light 665 nm), thinking or processing state (e.g., blue light 470 nm), or other general operational states. This capability is beneficial because it can limit the amount of information that needs to be displayed on the main display **300** and allows a human, another robot, or a machine to receive information from the humanoid robot **100**, even when the human, robot, or machine is positioned directly to one side of the humanoid robot **100** (a position from which the human, robot, or machine could not see the main display **300**). Also, the light emitters **264**.**2***a*-**264**.**2***d* use less battery power than the display **300** and may be able to relay information more quickly to the human, robot, or machine. Alternatively, the indicator lights can signal an operator to immediately take note of a more complex condition or set of information that is comprehensively displayed on the display **300**, thereby ensuring that an operator properly assesses that complex condition or information pertaining to the humanoid robot **100**. It should be understood that in other embodiments, the illumination assembly may: (i) emit a light that surrounds the periphery of the frontal shell **228**, (ii) emit a light that surrounds the rear edge of the frontal shell **228**, or (iii) include one or more emitters positioned in other parts of the humanoid robot **100** (e.g., torso **204**, knee **406***a*/**406***b*, leg assembly **402***a*/**402***b*, arm assembly **208***a*/**208***b*, hand **216***a*/**216***b*, etc.).
B. Robot Architecture
(88) The humanoid robot **100** is designed to possess substantial similarities in form factor and anatomy to human beings, including many of the same major appendages that human beings have. The humanoid robot **100** includes an upper region **200**, a lower region **400** spaced apart from the upper region **200**, and a central region **600** that interconnects the upper region **200** and the lower region **400**. The humanoid robot **100** is depicted in FIGS. **1** and **2** in an upright, standing position P**1**, where a pair of feet **412***a*, **412***b* of the lower region **400** are standing on a floor or ground surface G, such that the lower region **400** supports the upper region **200** and the central region **600** above the floor or ground surface G.
(89) The upper region **200** includes the following principal parts: (a) the previously mentioned head

and neck assembly **202**, (b) a torso **204**, (c) left and right shoulders **206***a*, **206***b*, and (d) left and right arm assemblies **208***a*, **208***b*, each of which includes: (e) a humerus **210***a*, **210***b*, (f) a forearm **212***a*, **212***b*, (g) a wrist **214***a*, **214***b*, and (h) a hand **216***a*, **216***b*. The lower region **400** includes left and right leg assemblies **402***a*, **402***b*, each of which includes: (a) a thigh **404***a*, **404***b*, (b) a knee **406***a*, **406***b*, (c) a shin **408***a*, **408***b*, (d) an ankle **410***a*, **410***b*, and (e) a foot **412***a*, **412***b*. The central region **600** is located generally in, or effectively provides, a pelvis region for the humanoid robot **100**. Each of the components of the upper region **200** and the lower region **400** noted above includes at least one actuator, which is configured to move these components relative to one another. The central region **600** is also configured to allow movement of the upper and lower regions **200**, **400** relative to one another in a three-dimensional manner.

C. Head and Neck Assembly

(90) As shown in FIG. **1**, it can be observed that the head and neck assembly **202** of the humanoid robot **100** extends from the torso **204** and is designed to: (i) provide the humanoid robot **100** with a recognizably humanoid shape, (ii) house and protect one or more electronic components (e.g., the display **300**, light emitting assemblies **264***a*-**264***d*, sensors, and/or head actuators), and (iii) provide information to a nearby human using the display **300** and/or the light emitting assemblies **264***a*-**264***d*. As detailed in FIGS. **7-30**, the head and neck assembly **202** comprises: (i) a head portion **202***a*, and (ii) a neck portion **202***b* that is coupled to a lower end or extent of said head portion **202***a*.

(91) 1. Exterior Head Shape

(92) As shown in FIGS. **1-9**, the head portion **202***a* of the humanoid robot **100** has an exterior surface SE that provides said head portion **202***a* with an overall shape that is similar to the shape of a human head. This overall shape of the head portion **202***a* is primarily defined by a housing assembly **220**, which includes: (i) a rear shell, rear head covering, or rear cover **234**, and (ii) the frontal shell, frontal shield, frontal head covering, or frontal cover **228**. In some embodiments, the head portion **202***a* is formed with no flat surfaces and is generally egg-shaped when viewed from the front, as shown in FIG. **4**, and the top, as shown in FIG. **7**. The head portion **202***a* of the humanoid robot **100** changes constantly in width from its top to its bottom, wherein the width of the head portion **202***a* increases from a top or scalp end **235** to a temple region **350**, where the head portion **202***a* is at its widest. The temple region **350** generally corresponds to the eye level of a human, or is situated at a location that is approximately 30-50% of a height **240** of the head portion **202***a* as measured from the top end **235**. The width of the head portion **202***a* then decreases from the temple region **350** to a lower or chin end **239**. In this manner, the head portion **202***a* of the humanoid robot **100** is asymmetrical about a first plane **102** that passes through a center or centroid C of the head portion **202***a* and is parallel with the transverse plane, equidistant from the top end **235** and the lower end **239**. The head portion **202***a* of the humanoid robot **100** is, however, symmetrical about a second plane **104** that is perpendicular to the first plane **102** passes through the center or centroid C of the head portion **202***a*, and is parallel with the sagittal plane. In other embodiments, it is conceivable that the head portion **202***a* may be symmetrical about the first plane **102** and asymmetrical about the second plane **104**.

(93) As shown in FIG. **4**, the rear shell **234** of the housing assembly **220** has a maximum height **240**, and the frontal shell **228** of the housing assembly **220** has a second height **242** that is less than the first height **240**. The rear shell **234** also has a first maximum width **244**, and the frontal shell **228** has a second width **246** that is less than the maximum width **244**. The head portion **202***a* of the humanoid robot **100** has the overall maximum height **240** that is greater than the maximum width **244**. The maximum height **240** and maximum width **244** of the head portion **202***a* are both provided by the rear shell **234**. The maximum width **246** of the frontal shell **228** is located at a position that is above the location of the maximum width **244** of the head portion **202***a*.

(94) As illustrated in FIG. **5**, a depth of the head portion **202***a* of the humanoid robot **100** is defined by a combination of both the rear shell **234** and the frontal shell **228**. This depth includes a

maximum depth **250** at a location that is approximately equal to the temple region **350**, and this maximum depth **250** extends between a front or facial region **241** of the head portion **202***a* to an occipital region **359** of the head portion **202***a*. The front end **241** of the head portion **202***a* is provided by the frontal shell **228**, and the rear end **243** is provided by the rear shell **234**. The depth of the head portion **202***a* changes constantly from the top end **235** to the lower end **239**. Specifically, the depth increases from the top end **235** to the maximum depth **250** and then decreases from the maximum depth **250** to the lower end **239**. The head portion **202***a* of the humanoid robot **100** is asymmetrical about a third plane **106** that extends through the center or centroid C of the head portion **202***a*, is perpendicular to the first plane **102**, and is parallel with the coronal plane. The head portion **202***a* of the humanoid robot **100** is also asymmetrical about first plane **102**. In other words, in this described embodiment, the head portion **202***a* of the humanoid robot **100** is only symmetrical about the second plane **104**. The center C is defined as being spaced at equal distances from: (i) the top end **235** and the bottom end **239**, (ii) the front end **241** and the rear end **243**, and (iii) lateral sides **247**, **249** of the head portion **202***a*. In other embodiments, the head portion **202***a* may be symmetrical about the first plane **102** and/or symmetrical about the third plane **106**. Stated another way, other embodiments of the head portion **202***a* may be symmetrical about: (i) all planes **102**, **104**, and **106**, (ii) two of the four planes **102**, **104**, and **106**, (iii) one of the four planes **102**, **104**, and **106**, or (iv) none of the planes **102**, **104**, and **106**.
(95) Exterior surfaces SE of the rear shell **234** and the frontal shell **228** are concave relative to: (i) the center C of the head portion **202***a* and/or (ii) the rear shell **234**. The head portion **202***a* of the humanoid robot **100** may include a nape region **245** formed in the rear extent of the head portion **202***a*, positioned below the occipital region **359** and above the neck portion **202***b*. The nape region **245** may include an exterior surface that is convex relative to the center C of the head portion **202***a*. In some embodiments, the nape region **245** is the only region of the head portion **202***a* that has a convex surface relative to the center or centroid C of the head portion **202***a*.
(96) When viewed from the front, as shown in FIGS. **4** and **14**, (i) the lateral sides of the head portion **202***a* exhibit a first vertical curvature or middle curvature generally extending about the center or centroid C and from a third point **221.2** to a second point **221.4**; (ii) the top extent **235** of the head portion **202***a* has a second vertical curvature or an upper curvature generally extending about the center or centroid C and from a second point **221.4** to a first point **221.6**; and (iii) the bottom end **239** has a third curvature or lower curvature generally extending about the center or centroid C and from third point **221.2** to fourth point **221.8**. The second curvature is less than, or has a lesser degree than, the first curvature. The second curvature or upper curvature is greater than, or has a greater degree than, the third curvature or the lower curvature. The first curvature or middle curvature is defined between third point **221.2** located at an upper extent of the buccal region **352** and the second point **221.4** located at a lower extent of the crown region **270** of the head portion **202***a*. The second curvature is primarily defined by the crown region **270**, wherein said curvature extends from the second point **221.4** located at a lower extent of the crown region **270** and the first or top point **221.6** located at the apex of the housing assembly **220**. The third curvature is defined between the third point **221.2** at the uppermost extent of the buccal region **352** and fourth point **221.8** at the lowermost extent of the buccal region **352**. The head portion **202***a* has a substantially oval shape when viewed from above, as shown in FIG. **7**, but may taper inwardly slightly in the frontal shell **228** towards the facial region **241** provided by this frontal shell **228**.
(97) 2. Head Housing Assembly
(98) The head housing assembly **220** of the head and neck assembly **202** is configured to contain and protect other assemblies that are housed within the head portion **202***a*. As discussed above, the housing assembly **220** is configured to have a form resembling the general shape of a human head and includes: (i) a frontal shell, frontal shield, frontal head covering, or frontal cover **228**, (ii) a rear shell, rear head covering, or rear cover **234**, (iii) an intermediate cover, intermediate support, or intermediate member **252**, and (iv) an electronics support or frame **254**. As discussed below, the

intermediate cover **252** and the electronics support **254** may be combined into a single integrated structure. Additionally, in other embodiments, the intermediate cover **252** may be omitted, and the electronics support **254** may then be directly coupled to an extent of the rear shell **234**. In further alternative embodiments, the electronics support **254** may be omitted, while the intermediate cover **252** is retained. Also, the rear shell **234** may be omitted or substantially omitted and effectively replaced by a substantially larger frontal shell **228**. Finally, conversely, the frontal shell **228** may be omitted or substantially omitted and replaced by a substantially larger rear shell **234**.
(99) The intermediate cover **252** and the rear shell **234** are designed to mount to one another and together define a first head sub-volume **236** within the housing assembly **220**. This first head sub-volume **236** is configured to contain and protect one or more components utilized in the operation of the humanoid robot **100**, such as various electronics, batteries, computing components, and the like. The frontal shell **228** provides the front end of the housing assembly **220** and defines a second sub-volume **238**, located between the intermediate cover **252** and the frontal shell **228**, also within the housing assembly **220**. This second sub-volume **238** is separated from the first sub-volume **236** by the intermediate cover **252** and is configured to contain and protect one or more components included in an electronics assembly **222**, such as the display **300**, light emitters **264**.**2***a*-**264**.**2***d*, cameras, etc. Both The frontal shell **228** and/or the intermediate cover **252** can be removed from the rest of the housing assembly **220** to service components within the sub-volumes **236**, **238** or to upgrade components located in these sub-volumes **236**, **238**. This modular design allows for individual components to be replaced without necessitating the replacement of the entire housing assembly **220**.
(100) i. Intermediate Cover
(101) The intermediate cover **252** includes structures that are used to mount components of the electronics assembly **222** to the head portion **202***a*. The intermediate cover **252** is configured to couple with the rear shell **234** and is located between the first and second sub-volumes **236**, **238** to separate these first and second sub-volumes **236**, **238**. In other words, the intermediate cover **252** is designed to split or divide the first sub-volume **236** from the second sub-volume **238**. In alternative embodiments, the intermediate cover **252** may be omitted entirely, in which case the first and second sub-volumes **236**, **238** may be converted into a single, larger sub-volume. Alternatively, the intermediate cover **252** may be combined or integrally formed with other structures disclosed herein (e.g., the electronics support **254**, the rear shell **234**, and/or the frontal shell **228**), whereby the first and second sub-volumes **236**, **238** may either remain distinct or may be combined into a single sub-volume depending on the specific integrated design. Further, it should be understood that other mounting structures, dividers, covers, and/or plates may be included within the head portion **202***a* to further sub-divide the housing assembly **220** into additional sub-volumes (e.g., potentially ranging from 3-10 sub-volumes).
(102) The intermediate cover **252** has an outer perimeter **256** that is sized to fit within an inset rim **258** of the rear shell **234**. In this manner, the outer perimeter **256** is slightly less than the inner perimeter of an extent of the rear shell **234** that is positioned between a ledge **257** and a forward edge **253** of an outer perimeter **259** of the rear shell **234**. As such, the outer perimeter **256** of the intermediate cover **252** has a length that is less than the length of the outer perimeter **259** of the rear shell **234**. In other embodiments, the outer perimeter **256** of the intermediate cover **252** may not be sized to fit within an inset rim **258** of the rear shell **234**. Instead, said outer perimeter **256** of the intermediate cover **252** may be substantially equal to the outer perimeter **259** of the rear shell **234**, wherein said intermediate cover **252** may be coupled to, positioned adjacent to, and/or abutting said forward edge **253** of the rear shell **234**. In other words, a rear extent of the intermediate cover **252** may be configured to abut the forwardmost surface of said forward edge **253** of the rear shell **234**. In yet other embodiments, the outer perimeter **256** of the intermediate cover **252** may only extend along an extent that is less than substantially all, or even less than a majority (e.g., along only a minority), of the inner perimeter of an extent of the rear shell **234** that is positioned between a

ledge **257** and a forward edge **253** of the outer perimeter **259**.
(103) The outer perimeter **256** of the intermediate cover **252** is also slightly less than an outer perimeter **260** of the frontal shell **228**. As such, the outer perimeter **256** of the intermediate cover **252** has a length that is less than the length of the outer perimeter **260** of the frontal shell **228**. In other embodiments, the outer perimeter **256** of the intermediate cover **252** may be substantially equal to the outer perimeter **260** of the frontal shell **228**. In further alternative embodiments, the outer perimeter **256** of the intermediate cover **252** may only extend along an extent that is less than substantially all, or even less than a majority (e.g., along only a minority), of a rear edge **322** of the frontal shell **228**.
(104) The intermediate cover **252** further includes a plurality of peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d*, which are spaced around the outer perimeter **256** of the intermediate cover **252**. Said peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d* form a plurality of light emitter housings **262.10***a*-**262.10***b*. Specifically, each of these peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d* is configured to house a respective light emitting assembly **264***a*, **264***b*, **264***c*, **264***d* of the illumination assembly **263**. Said light emitter housings **262.10***a*-**262.10***b* have five primary walls that are comprised of: (i) two end walls **262.10.2**, **262.10.4**, wherein said end walls **262.10.2**, **262.10.4** are angled (e.g., at an obtuse angle) relative to a frontal surface SIF of the intermediate cover **252**; (ii) a top wall **262.10.6**; (iii) a bottom wall **262.10.8**, wherein said bottom wall **262.10.8** is also angled relative to the frontal surface SIF of the intermediate cover **252**; and (iv) an interior wall **262.10.10**, wherein said interior wall **262.10.10** is angled (e.g., at an obtuse angle) relative to the frontal surface SIF of the intermediate cover **252**. The angled configuration of these two end walls **262.10.2**, **262.10.4**, the bottom wall **262.10.8**, and the interior wall **262.10.10** is designed to achieve two objectives: (i) to effectively direct light out of the housings **262.10***a*-**262.10***b*, and (ii) to ensure that the light emitted radiates through a diffuser or lens **264.4***a*-**264.4***d* in a manner that does not let the light scatter too broadly, nor does it overly restrict the scattering of the light. In other embodiments, the walls **262.10.2**, **262.10.4**, **262.10.8**, **262.10.10** may not be angled relative to the frontal surface SIF of the intermediate cover **252**, the top wall **262.10.6** may be angled relative to the frontal surface SIF of the intermediate cover **252**, and/or the angles between the walls **262.10.2**, **262.10.4**, **262.10.8**, **262.10.10** and the frontal surface SIF of the intermediate cover **252** may be an acute angle instead of obtuse angle.
(105) ii. Electronics Support
(106) The electronics support **254** is mounted to a generally central area of the intermediate cover **252** and is configured to position the display **300**, which is included in the electronics assembly **222**, between the intermediate cover **252** and the frontal shell **228**, specifically within the second head sub-volume **238**. The electronics support **254** includes a base coupling **266** configured to mount to the intermediate cover **252**, and a display coupling **268** configured to mount the display **300** to the electronics support **254**. The base coupling **266** is located above and rearward of the display coupling **268**. The display coupling **268** positions the display **300** in a spaced apart relation to the frontal shell **228**, as clearly shown in FIG. **15**. As shown in FIGS. **26** and **27**, the intermediate cover **252** is configured to lie behind the electronics assembly **222** to locate at least a portion of the electronics assembly **222** between the intermediate cover **252** and the frontal shell **228**.
(107) iii. Frontal Shell
(108) The frontal shell **228** is configured to cover at least the intermediate cover **252** and the electronics assembly **222**, as shown in FIGS. **1-27**. The frontal shell **228** forms a forwardmost, exterior surface of the head portion **202***a* and cooperates with the intermediate cover **252** to define the second sub-volume **238** within the housing assembly **220**. The frontal shell **228** may be made from a transparent material so that the display **300**, which is mounted on the electronics support **254**, may be viewed therethrough. In other embodiments, the frontal shell **228** may be tinted or entirely opaque. The intermediate cover **252** can be similarly colored (tinted or opaque) so that the frontal shell **228** has a similar appearance to the intermediate cover **252**, and so that lights or

images displayed on the display **300** are the only items that are visible or displayed/conspicuous through the frontal shell **228**. The frontal shell **228** may be coated, etched, or formed in with a plurality of layers (e.g., examples of which are disclosed within U.S. Pat. Nos. 8,770,749, 9,134,547, 9,383,594, 9,575,335, 9,910,297, all of which are incorporated herein by reference) in a manner that improves durability, increases sensor accuracy, filters one or more specific wavelengths of light, reduces glare, enhances overall appearance, reduces fogging, makes the frontal shell **228** easier to clean or protects it from cleaning products. Examples of such optical coatings include anti-reflection coatings, mirror coatings, hard coatings, anti-static coatings, anti-fog coatings, some of which are described within U.S. patent application Ser. Nos. 16/896,016, 16/698,775, 16/417,311, 16/126,983, 15/359,317, 15/515,966, each of which are incorporated herein by reference. Further, the material composition, shape, number of layers, and composition of said layers of the frontal shell **228** may be different from the material composition, shape, number of layers, and composition of said layers utilized within other parts of the frontal shell **228** itself. In other words, the composition, shape, number of layers, and composition of said layers may vary across different regions of the frontal shell **228**. It should be understood that this disclosure is not limited to just the information that is disclosed within those specifically cited applications; but instead should encompass any compositions, shapes, layer numbers, and compositions of layers that are known in the art or are rendered obvious in light of what is currently known in the art.
(109) The frontal shell **228**, or an extent thereof, may have a substantially uniform thickness, which may be equal to or greater than 1 mm, and is preferably greater than 2 mm. Additionally, the frontal shell **228** may be optically correct and is typically not designed to be a corrective lens. As such, the frontal shell **228** has a dioptric power of less than 0.25 diopters, preferably less than 0.12 diopters, and most preferably less than 0.06 diopters. The frontal shell **228** may exhibit a reverse/negative pantoscopic tilt, a forward/positive pantoscopic tilt, or no pantoscopic tilt at all. Accordingly, the frontal shell **228** may be made from or may include materials such as polycarbonate (PC), acrylic (PMMA), trivex, nylon, gorilla glass (aluminosilicate glass), thermoplastic polyurethane (TPU), high-grade glass, cr-39, polyethylene terephthalate (PET), polystyrene, fused silica (quartz glass), borosilicate glass, polyurethane, cellulose acetate, polyvinyl chloride (PVC), cellulose acetate butyrate (CAB), polyvinyl butyral (PVB), optical-grade resin, sapphire glass, polyetherimide (PEI), Lexan, thermoset plastics, other anti-scratch coated plastics, or any other similar material that is known in the art.
(110) In the illustrative embodiment shown in FIGS. **1-16**, the frontal shell **228** is mounted to the intermediate cover **252** and/or the rear shell **234** along a shield interface **320**, which is located at the outer perimeters **256**, **260** of the rear shell **234** and the frontal shell **228**. A rear facing edge **322** of the frontal shell **228** is configured to mount with: (i) the forward edge **253** of the rear shell **234**, (ii) a frontal extent of the intermediate cover **252**, and/or (iii) both, to form the shield interface **320**. As shown in this embodiment, the shield interface **320** is not flat or planar. Instead, the shield interface **320** is irregular due to the fact that the rear edge **322** of the frontal shell **228** is formed to include a plurality of recesses **324**. Each recess **324** is sized to receive a respective one of the peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d* that are formed in the intermediate cover **252** and that house the respective light emitting assemblies **264***a*-**264***d*. In some embodiments, the frontal shell **228** itself could include the peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d*, although this configuration may not be desirable as the frontal shell **228** is the component that is most likely to be removed from the housing assembly **220** for servicing the head portion **202***a* and the electronics assembly **222**.
(111) Except for the aforementioned recesses **324**, the rear facing edge **322** of the frontal shell **228** is substantially planar along the interface **320**. The shield interface **320** between the intermediate cover **252** and the frontal shell **228**, and thus the rear-facing edge **322**, extends at an angle A**2** relative to the third plane **106**, as shown in FIGS. **5**, **15**, **31**, and **57**. In the illustrative embodiment, this angle A**2** is within a range of about 15 degrees to about 50 degrees, preferably between 20 and

40 degrees, most preferably between 25 and 35 degrees, and may be approximately 30 degrees. This angular relationship provides the frontal shell **228** with a larger depth at a top end thereof, which serves to increase the volume of the second sub-volume **238** and thereby provide more room for components of the electronics assembly **222**. An upper end of the frontal shell **228**, near an upper extent of the head portion **202***a*, is located rearward of the third plane **106**, while a lower end of the frontal shell **228**, near a chin region **355**, is located forward of the third plane **106**, as depicted in FIG. **5**A. There are no recesses formed in the frontal shell **228** in the orbital region **368**, the nasal region **357**, the oral region **366**, or the frontal region **362**. The frontal shell **228** extends upward from an extent of the rear shell **234** that is positioned in the chin region **355**, continuing over a majority of the facial region **241**, and into or beyond a frontal edge of the parietal region **360**. The frontal shield **228** has an outer surface that occupies at least an orbital region **368** and a nasal region **357**. Significantly, The orbital region **368** of the frontal shell **228** is not recessed in comparison to the nasal region **357** of the frontal shell **228**.

(112) The described depth change of the frontal shell **228** positions a first light emitting assembly **264***a* and a third light emitting assembly **264***c* above and rearward of a second light emitting assembly **264***b* and a fourth light emitting assembly **264***d*. Such an arrangement provides a greater viewing area for users to observe at least one light emitting assembly **264***a*-**264***d* when they are positioned at different orientations relative to the humanoid robot **100**. In other embodiments, the first and third light emitting assemblies **264***a*, **264***c* may not be positioned rearward of the second and/or fourth light emitting assemblies **264***b*, **264***d*. Instead, the second and/or fourth light emitting assemblies **264***b*, **264***d* may be positioned in the same vertical plane as the first and third, and/or may even be positioned rearward of the first and third light emitting assemblies **264***a*, **264***c*.

(113) The frontal shell **228** may: (i) wrap from the front of the head portion **202***a* into the side regions of the head portion **202***a*, (ii) extend into the chin area region **355** or cover the entire chin area, (iii) may have a non-uniform rear edge **322**, which is formed by the plurality of recesses **324**. The plurality of recesses **324** may be configured to receive an extent of the peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d*.

(114) The frontal shell **228** may not extend to the crown region **270** of the head portion **202***a* and/or may not extend rearward past a location where a human's ears would typically be located. The disclosed frontal shell **228** may occupy between 25% and 50% of the surface area of the head portion **202***a* and may be curved in at least two directions (e.g., vertically and horizontally). In some embodiments, the frontal shell **228** and the display **300** may be integrated into a single component or may be formed from a plurality of interconnected components. It is also possible that The frontal shell **228** may have a different curvature than the display **300**.

(115) As shown in FIG. **14**, the frontal shell **228** has in a forward facing orientation OFF, at least: (i) a substantially horizontal first arc length AL.sub.1 or chin arc length that extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a first location or horizontal plane **120**; (ii) a substantially horizontal second arc length AL.sub.2 or a display arc length hat extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a second location or horizontal plane **122**; (iii) a substantially horizontal third arc length AL.sub.3 or forehead arc length, or above the display arc length that extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a third location or horizontal plane **124**; and (iv) a substantially horizontal fourth arc length AL.sub.4 or crown arc length that extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a fourth location or horizontal plane **126**.

(116) The frontal shell **228** is concave relative to the display **300** at each of these locations **120**, **122**, **124**, **126**, such that the frontal shell **228** extends at least partially about the display **300**. The first arc length occurs at a position below the display **300**. The second arc length occurs at approximately the center C of the head portion **202***a*, is aligned with the display **300**, and is greater in magnitude than the first arc length. The third arc length occurs at a position above the display

300 and is greater in magnitude than the second arc length. The fourth arc length occurs at a position above the third arc length and also above the display **300**, and is less in magnitude than both the second and third arc lengths.

(117) As shown in FIG. **4**, the frontal shell **228** also has, at least: (i) a first width W.sub.1 or lower frontal shell width that extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a first location below the first horizontal plane **120**; (ii) a second width W.sub.2 or frontal shell centroid width that extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a second location or at the horizontal plane **102** (which passes through the center C); and (iii) a third width W.sub.3 or upper frontal shell width that extends from a first edge extent **323***a* of the rear edge **322** to a second edge extent **323***b* of the rear edge **322** at a third location above the first plane **102**. The first width W.sub.1 is less than both the second width W.sub.2 and the third width W.sub.3; the second width W.sub.2 is greater than both the first width W.sub.1 and the third width W.sub.3; and the third width W.sub.3 is greater than the first width W.sub.1 but less than the second width W.sub.2. The first width W.sub.1 or lower frontal shell width is positioned below a display width W.sub.D, and wherein the display width W.sub.D is greater than the frontal shell width W.sub.1. The frontal shell centroid width W.sub.2 and the upper frontal shell width W.sub.3 is greater than the display width W.sub.D.

(118) iv. Rear Shell

(119) The rear shell **234** is shaped to resemble the curvature of a rear and sides of a human head, or at least portions of a parietal region **360**, an occipital region **359**, a temporal region **350**, an auricular region **356**, a zygomatic region **358**, a mastoid region **364**, a buccal region **352**, and a parotid region **354**. As shown in FIGS. **5**B and **15**B, the interface region **320** between the rear shell **234** and the frontal shell **228** passes through the buccal region **352**, the parotid region **354**, the zygomatic region **358**, the temporal region **350**, the parietal region **360**, and the crown region **270**. In other words, the rear shell **234** begins at the parotid region **354**, the zygomatic region **358**, the temporal region **350**, the parietal region **360**, and the crown region **270** and forms all portions of the head portion **202***a* rearward thereof. Likewise, the frontal shell **228** begins at the parotid region **354**, the zygomatic region **358**, the temporal region **350**, the parietal region **360**, and the crown region **270** and forms all portions of the head portion **202***a* forward thereof except for a chin region **355** which is included as part of the rear shell **234**. Also, it should be understood that in some variations, the frontal shell **228** may only be positioned forward of the auricular region **356**.

(120) The rear shell **234** is configured to cover a rear portion of the electronics assembly **222** and to form the rear end of the head portion **202***a*. The rear shell **234** extends downward from a top central position and also forward at an angle that is substantially similar to that of the rear facing edge **322** of the frontal shell **228**. The rear shell **234** includes a forward facing edge **298** that is configured to mate with the rear facing edge **322** of the frontal shell **228** and/or with the intermediate cover **252**.

(121) As shown in FIG. **16**, the rear shell **234** has an outer rim **255** and a ledge **257** that projects inwardly from this outer rim **255** and is located within the first sub-volume **236**. The rim **255** may have a slightly larger outer perimeter **259** than the outer perimeter **256** of the intermediate cover **252**; this sizing allows the intermediate cover **252** to fit snugly within the rim **255** and engage the ledge **257**. Both The ledge **257** and the intermediate cover **252** have corresponding attachment holes **261***a*, **261***b* that are designed to receive a fastener to securely mount the intermediate cover **252** to the rear shell **234**. Apertures **261***c*, which are formed in the frontal shell **228**, also receive a fastener to mount the frontal shell **228** to both the intermediate cover **252** and the rear shell **234**.

(122) The rear shell **234** further includes a chin projection region **269** that extends forward from the rim **255** and defines a lower end of the rear shell **234**. The intermediate cover **252** further includes a lower mount **265** that engages and rests upon this chin projection region **269**. The chin projection region **269** and the lower mount **265** each include corresponding attachment holes **267***a*, **267***b* that receive a fastener to secure the intermediate cover **252** to the rear shell **234** at a lower end thereof. Apertures **267***c*, formed in the frontal shell **228**, also receive a fastener to mount the frontal shell

228 to both the intermediate cover **252** and the rear shell **234** at this lower junction.
(123) The rear shell **234** may be fabricated from a variety of materials, including but not limited to: silicone elastomers, thermoplastic polyurethane (TPU), shape-memory polymers (SMPS), polydimethylsiloxane (PDMS), polyurethane, liquid silicone rubber (LSR), urethane rubber, vinyl (PVC) skin, soft thermoplastic elastomers (TPE), elastomeric alloys, acrylonitrile butadiene styrene (ABS) blends, high-density polyethylene (HDPE) blends, conductive polymers, carbon nanotube-infused elastomers, magnetic shape-memory alloys, electroactive polymers (EAPS), styrene-butadiene rubber (SBR), thermoplastic vulcanizates (TPV), polyurea elastomers, medical-grade synthetic skin materials, thermoplastic olefins (TPO), fluoroelastomers, chloroprene rubber, ethylene propylene diene monomer (EPDM) rubber, polyacrylamide hydrogels, polycaprolactone (PCL), photocurable resins, elastomeric composites, phosphorescent elastomers, thermochromic materials, electrostrictive polymers, piezoelectric polymers, superelastic alloys, microcellular foams, hyperelastic materials, viscoelastic gels, nanocomposite elastomers, fabrics, metal, other similar plastics or polymers, any combination of the above, and/or any other similar material known in the art. The rear shell **234** may be manufactured using any known method, including techniques such as: molding (e.g., injection molding or dip molding), casting, 3D printing (additive manufacturing), dip molding and subsequent coating, spray coating, lamination and layering, electrospinning, sculpting and machining, thermoforming, any combination of these methods, and/or any other known manufacturing method.
(124) 3. Electronics Assembly
(125) The electronics assembly **222**, which is contained within the head portion **202***a*, may include one or more of the following components: (i) a sensor assembly **301***a*, (ii) the display **300**, (iii) a directional microphone, (iv) one or more speakers, (v) antennas, (vi) the illumination assembly that includes at least one, and preferably a plurality of, light emitting assemblies **264***a*-**264***d*, (vii) a data storage device, and (viii) other miscellaneous electronics (e.g., an Inertial Measurement Unit (IMU), an RFID reader, location sensors (such as Global Positioning System ("GPS"), GLONASS, Galileo, QZSS, and/or iBeacon technology), etc.), and/or Printed Circuit Boards (PCBs) for connecting these various electronic components. The data storage device may be a removable memory device or may be integrated within a computing device that comprises a processor and associated memory. In some examples, the data storage device may be housed in another portion of the humanoid robot **100**, such as within the torso **204**. In some examples, the data storage device may be configured to store data that is collected from other components of the humanoid robot **100**.
(126) As shown in at least FIGS. **19-20**, the components of the electronics assembly **222** may be mounted to the electronics support **254** and located above the display **300**; alternatively, they may be mounted to an internal mounting frame that supports the head and neck assembly **202** on the torso **204**, or to any other suitable structure within the head and neck assembly **202**. Mounting the electronics assembly **222** to the electronics support **254** above the display **300** and within the second sub-volume **238** may allow for the individual items to be arranged in a space-saving manner and may leave other areas of the head and neck assembly open for the storage of other components that do not require direct access to areas outside of the humanoid robot **100**, such as sensors, cameras, displays, etc. As noted previously, the housing assembly **220** is configured to enclose the electronics assembly **222** without interfering with the transmission or reception of signals. For example, the housing assembly **220** does not obscure the line of sight of the sensors.
(127) i. Display
(128) As best shown in FIGS. **17-25**, the display **300** of the electronics assembly **222** may be mounted to the electronics support **254** and positioned behind the frontal shell **228**. The display **300** is operatively connected to at least one processor to generate and display status messages and other types of information on the display **300**. For example, the display **300** may display information: (i) related to the robot's operational state (e.g., working, error, moving, etc.); (ii) obtained from sensors contained within the head portion **202***a*, or (iii) received from other processors that are in

communication with the display **300** (e.g., other internal processors housed within the humanoid robot **100** or external information transmitted to and received by the humanoid robot **100**)). The information may be displayed in the format of blocks, well-known shapes, logos, or other moving items (e.g., animated thought bubbles). However, as stated before, the information may not be displayed in connection with human facial features (e.g., eyes, mouth, nose).

(129) As shown in FIGS. **17-25**, the display **300** may have a substantially rectangular display surface that has a convex curvature that conforms with the curvature of the frontal shell **228** of the housing assembly **220**. The display **300** has a first or left substantially vertical edge **300.2** and a second or right substantially vertical edge **300.4** when the head is in the forward facing orientation OFF, and wherein said display **300** also includes a display width W.sub.D that extends between the first or left substantially vertical edge **300.2** and the second or right substantially vertical edge **300.4**. The display width W.sub.D is greater than the first width W.sub.1 or lower frontal shell width, but is less than the frontal shell centroid width W.sub.2 and the upper frontal shell width W.sub.3. The display **300** also includes an upper edge **300.6** and a lower edge **300.8**, wherein the display **300** has a constant: (i) height H between the upper edge **300.6** and the lower edge **300.8** across the display width W.sub.D, and (ii) a constant arc length between the upper edge **300.6** and the lower edge **300.8**.

(130) The display **300** may also be slightly tilted downward. For example, the display **300** may be tilted from a horizontal plane at an angle of about 5.7 degrees to about 8.6 degrees, or more specifically, about 6.4 degrees to about 7.9 degrees. This tilted orientation of the display **300** increases viewability for an observer and helps to eliminate unwanted reflections. The display **300** may use any known display technology or feature including, but not limited to: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LEC, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay, resistive touchdisplay, monochrome, color, or any combination of the above, or any other known technology or display feature. The displays disclosed herein meet the standards described in FDA CFR Title 21 part **1040.10**, titled Performance standards for Light-Emitting Products, and ANSI LIA Z136.1, titled Safe Use of Lasers, at the time of filing this application, both of which are incorporated herein by reference.

(131) It should be understood that this application contemplates the use of displays **300** that have different sizes. To this end, the display **300** may extend between any two of the horizontal lines shown in FIGS. **22** and **23**. For example, the display **300** may extend from the third line from the bottom to the third line from the top in these figures. In other examples, the display **300** may be sized to fit between any two of the lines shown in FIGS. **24** and **25**. Additionally, each of the lines indicated on the display **300** in FIGS. **22-24** can represent different zones or segments included in the display **300**, which can be used to convey different images or other visual representations across the surface of the display **300**. As an example, the sides of the display **300** can be used to display a different image or visual representation compared to a front portion of the display **300**. Alternative display sizes may be used for various reasons, such as: (i) to reduce the surface area of fragile elements within the humanoid robot **100**, (ii) because the humanoid robot **100** is not primarily designed to work in close proximity to humans, (iii) because additional area within the head portion **202***a* is needed for sensors or other electronics, or (iv) for any other reason that would be known or apparent to one of skill in the art.

(132) The disclosed display **300** may be embedded in or occupy the entire frontal shell **228**, between 100% and 75% of the frontal shell **228**, between 75% and 50% of the frontal shell **228**, between 50% and 25% of the frontal shell **228**, or less than 25% of the frontal shell **228**. In some examples, the display **300** may utilize the full area of the frontal shell **228**. In some examples, the display **300** may be sized to fit between any two of the lines shown in FIGS. **50-51**. In some further examples, the display **300** may be sized to fit between any two of the lines shown in FIGS. **52-53**. In other words, the ratio of the display **300** size to the frontal shell **228** size may be any suitable

ratio. The display **300** may be curved in a single direction, in two directions (e.g., vertically and horizontally), or it may embody a freeform design that could include multiple complex curves. In certain embodiments, the frontal shell **228** and the display **300** may be integrated into a single, cohesive unit. As best shown in FIGS. **15**A, a gap is formed between the outer surface of the display **300** and the inner surface of the frontal shell **228**.

(133) As shown in FIG. **15**B, the display **300** is curved so as to occupy and/or coincide with and display information through all or portions of the orbital region **368**, the frontal region **362**, the temporal region **350**, the zygomatic region **358**, the nasal region **357**, the infraorbital region **363**, the buccal region **352**, and the oral region **366** of the head portion **202***a*. The display **300** may not coincide with any region located rearward of the shield interface **320**, including a mental region **355**, an auricular region **356**, a crown region **270**, a parietal region **360**, an occipital region **359**, or a mastoid region **364**. In other embodiments, the display **300** may present information, indications, or visual representations across: (i) the entire frontal shell **228**, (ii) a majority of the frontal shell **228**, (iii) the entire facial region **241**, and/or (iv) a majority of the facial region **241**. In contrast, in other alternative embodiments, the display **300** may be omitted entirely or moved to a different region (e.g., the torso **204**) of the humanoid robot **100**.

(134) FIGS. **9**A-**9**G show various statuses and their corresponding indications or visual representations that are contemplated by this disclosure for display on the display **300**. It should be noted that the present disclosure is not limited to these specific statuses and their corresponding indications or visual representations and that these are merely examples that can be displayed by the display **300**.

(135) FIGS. **8** and **9**A show sets of information displayed by the display **300**, including device status (e.g. camera status and robot start-up status), robot part status (e.g. a specified joint status), and a battery level status. The various statuses disclosed herein can update as the actual status changes. For example, during the start-up sequence of a camera(s) **302**, **303**, **304**, the display **300** may display text, an icon, or another visual representation indicating that the camera(s) **302**, **303**, **304** is currently initializing and is not yet ready for use. The status can then change to display an icon, or another visual representation, indicating that the camera(s) **302**, **303**, **304** are active once the start-up process is complete.

(136) FIG. **9**B shows an icon indicating a battery status, specifically that the battery is charging. The icon is generally in the shape of a battery. The icon further includes a lightning bolt symbol to indicate that the battery is currently in a charging state. This icon (or icons) can be displayed in a first color (e.g. green) while the battery is charging and/or after the battery has been fully charged. FIG. **9**C shows another icon indicating an additional battery status, in this case, a low battery level. This icon is also generally in the shape of a battery and includes a level marker to indicate the current battery charge level. In the illustrative embodiment of FIG. **9**C, the level marker is depicted as a thin line to one side of the battery icon, visually indicating that the battery charge is low and that the battery should be recharged soon. This level marker can move or increase in size to indicate the current battery charge level at any given time.

(137) FIG. **9**D shows another icon indicating an alert or a system failure event. The icon includes a triangle with an exclamation mark contained within it to signify the alert or system failure event. Such an icon may be displayed when a part of the humanoid robot **100** has failed, such as an actuator, a camera, or another device included in the humanoid robot **100**. Identifying information, in the form of text or another accompanying icon, can also be displayed along with the alert or failure icon to specify which particular part of the humanoid robot **100** has failed or requires attention. FIG. **9**E shows an icon indicating a particular operational mode of the robot. Illustratively, the mode depicted by this icon indicates a “follow mode,” in which the humanoid robot **100** is engaged in following a user, another device, or another robot. The icon includes a generally human-shaped figure with a border around this generally human-shaped figure. Other icons corresponding to additional modes of operation of the robot can also be displayed as needed.

(138) FIG. **9**F shows an icon indicating a particular task currently being performed by the humanoid robot **100**. Illustratively, the task icon depicted is a generally human shaped figure that is shown lifting and carrying a box or another item. Other icons corresponding to additional tasks that can be completed by the robot can also be displayed. FIG. **9**G shows an additional robot status icon. Illustratively, the status icon depicted is a “pause” icon. The pause icon can be displayed when the humanoid robot **100** is currently not completing any tasks or modes and is essentially in a ready state, awaiting instructions. Other icons corresponding to additional robot statuses that can be conveyed by the robot can also be displayed.
(139) ii. Head Illumination Assembly
(140) The head illumination assembly **263** includes at least one, and preferably a plurality of, light emitting assemblies **264***a*-**264***d* that are located on the lateral sides of the head portion **202***a*. In certain configurations, this illumination assembly **263** may be designed to visually indicate various robot statuses to users who are viewing the humanoid robot **100** from the side. As shown in FIGS. **5**B, **15**B, **31**B, and **57**B, a first light emitting assembly **264***a* is located in the temporal region **350** of the robot's head portion **202***a*, and a second light emitting assembly **264***b* is located in the buccal region **352** of the robot's head portion **202***a*. Third and fourth light emitting assemblies **264***c*, **264***d* are located symmetrically on the opposite side of the robot's head portion **202***a*, and so are also located in the temporal region **350** and the buccal region **352**, respectively. In some embodiments, the light emitting assemblies **264***a*-**264***d* can be located all or partially in a parotid region **354** of the robot's head portion **202***a*, an auricular region **356** of the robot's head portion **202***a*, a zygomatic region **358** of the robot's head portion **202***a*, a parietal region **360** of the robot's head portion **202***a*, a frontal region **362** of the robot's head portion **202***a*, or a mastoid region **364** of the robot's head portion **202***a*, so long as the light emitters **264**.**2***a*-**264**.**2***d* of the assemblies **264***a*-**264***d* are positioned on a lateral side of the robot's head portion **202***a* so as to be visible to a person standing next to the humanoid robot **100**. These strategic positions of the light emitting assemblies **264***a*-**264***d* allow users to view the light emitted from said light emitting assemblies **264***a*-**264***d* from the side while the humanoid robot **100** is, for example, working on a task in an assembly line, and while the main display **300** is facing the assembly line and may not be entirely visible by the user. Further, the light emitting assemblies **264***a*-**264***d* may face away from the display **300** so as not to obstruct the information being shown by the display **300**, and they may also face away from other sensors so as not to interfere with the operation of those sensors. Other regions of the head portion **202***a* where the light emitting assemblies **264***a*-**264***d* are not typically found include the chin or mental region **355**, the orbital region **368**, the nasal region **357**, the crown region **270**, and the occipital region **359**.
(141) The light emitting assemblies **264***a*-**264***d* in the head portion **202***a* may be configured to display a status of the humanoid robot **100**, or a part thereof, to users. For example, the light emitting assemblies **264***a*-**264***d* can display: (i) a first color (i.e., green) when the robot is engaged in a task, such as assembling a part on an assembly line, (ii) a second color (i.e., yellow) when the humanoid robot **100** is not currently assigned to a task, thereby indicating to users that the humanoid robot **100** is available for a new task, and (iii) a third color (i.e. red) when the humanoid robot **100** is low on battery life and should be recharged. The light emitting assemblies **264***a*-**264***d* and/or the display **300** can also be used to indicate when a component in the head portion **202***a* and/or the neck portion **202***b*, such as an actuator, is malfunctioning and should be serviced.
(142) The light emitting assemblies **264***a*-**264***d* can also incorporate one or more display sequences in which the light emitting assemblies **264***a*-**264***d* are turned off and on, or the light emitted from said light emitting assemblies **264***a*-**264***d* changes, in a specific pattern or sequence to indicate various statuses. For example, the light emitting assemblies **264***a*-**264***d* can blink repeatedly to indicate that the humanoid robot **100** has lost communication with a host server or an external device, or that it is attempting to pair with or searching for a device or server to connect to. The light emitting assemblies **264***a*-**264***d* may coordinate their display output with the information

being displayed on the main display **300**. For example, the light emitting assemblies **264***a*-**264***d* can display a particular color that corresponds with the information currently displayed on the display **300**. If the humanoid robot **100** is running low on battery life, the light emitting assemblies **264***a*-**264***d* can display a red color while the display **300** concurrently displays a message and/or an icon that indicates that the battery is low.

(143) The light emitting assemblies **264***a*-**264***d* can also be synchronized with other devices included in the humanoid robot **100** as well. For example, the light emitting assemblies **264***a*-**264***d* can be operated in conjunction with a speaker and may change colors or blink as the humanoid robot **100** outputs an audible message. Light emitting assemblies **264***b* and **264***d* are positioned adjacent to an oral region **366** of the head portion **202***a* and can be operated independently of the light emitting assemblies **264***a*, **264***c* which are located above light emitting assemblies **264***b*, **264***d* and adjacent to an orbital region **368** of the head portion **202***a*.

(144) Each of the light emitting assemblies **264***a*-**264***d* in the head portion **202***a* typically includes: (i) a light source or light emitter **264**.**2***a*-**264**.**2***d*, and (ii) a diffuser lens **264**.**4***a*-**264**.**4***d* covering the light source **264**.**4***a*-**264**.**4***d*. The light source **264**.**2***a*-**264**.**2***d* and the diffuser lens **264**.**4***a*-**264**.**4***d* form a unit **264**.**6***a*-**264**.**6***d* that is inserted together into each respective peripheral protrusion **262***a*, **262***b*, **262***c*, **262***d* to couple the light emitting assemblies **264***a*-**264***d* to the head portion **202***a*. The light source or emitter **264**.**2***a*-**264**.**2***d* can include any known light emitter, including any one or more of the following: laser, LCD, LED (e.g., COB LED), OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay, resistive touchdisplay, monochrome emitter, color emitter, or any combination of the above, or any other known technology or light-emitting feature. It should be understood that in other embodiments, the above-disclosed light sources or emitters **264**.**2***a*-**264**.**2***d* and/or additional light emitters **264**.**2***a*-**264**.**2***d* may be formed in any desirable configuration or used with any other material, structure, or component to form the desirable light emitting assemblies **264***a*-**264***d*. Examples of said light emitting assemblies **264***a*-**264***d* that may be formed include configurations utilizing fiber optic cables, electroluminescent (EL) wire, laser diodes, neon tubes, cold cathode fluorescent lamps (CCFL), plasma tubes, phosphorescent strips, UV LED strips, infrared LED arrays, light guide panels (LGP), or edge-lit light panels. The light source or light emitter **264**.**2***a*-**264**.**2***d* may be made from a single emitter or a plurality of emitters (e.g., a number ranging between 2 and 1000 individual emitters). Said light source or light emitter **264**.**2***a*-**264**.**2***d* may be driven by an internal driver or an external driver located within another aspect of the electronics assembly **222**. Each of the light emitters **264**.**2***a*-**264**.**2***d* is positioned in an inner portion of each respective peripheral protrusion **262***a*-**262***d* (i.e., positioned toward the display **300**) and the diffuser lens **264**.**4***a*-**264**.**4***d* is positioned in front of the light emitter **264**.**2***a*-**264**.**2***d* so as to reside between a frontal extent of the light emitter **264**.**2***a*-**264**.**2***d* and an outermost edge **262**.**2***a*-**262**.**2***d* of each respective peripheral protrusion **262**, and/or an outermost edge or surface of the head portion **202***a* or frontal shield **228**. In some embodiments, the diffuser lens **264**.**4***a*-**264**.**4***d* can be omitted from the light emitting assemblies **264***a*-**264***d*. Each light emitter **264**.**2***a*-**264**.**2***d* is located forward of and adjacent to the forward facing edge **298** of the rear shell **234**, and also rearward of and adjacent to a portion of the rear facing edge **322** of the frontal shell **228** that defines each recess **324**. A rearmost edge of each light emitter is located rearward of the entire rear facing edge **322** of the frontal shell **228**. The light emitters **264**.**2***a*-**264**.**2***d* are located in voids **262**.**4***a*-**262**.**4***d* that are defined by each respective peripheral protrusion **262***a*-**262***d*, where these voids **262**.**4***a*-**262**.**4***d* are located between the frontal shell **228** and the rear shell **234**. In other embodiments, the light emitters **264**.**2***a*-**264**.**2***d* may not be formed directly in these voids **262**.**4***a*-**262**.**4***d*; and instead, said voids **262**.**4***a*-**262**.**4***d* may act as a reflector for light that is emitted from said light emitter or source. In other words, the light emitter may be positioned in the first head sub-volume **236**.

(145) A gap, region, or channel **370** is formed between the recesses **324** formed in the rear edge of

the frontal shell **228** and the rear shell **234**, and wherein light emitted from the illumination assembly **263** (comprising the light emitting assemblies **264***a*-**264***d*) is visible in a region that contains a gap or channel **370**. In other words, the region or the gap **370** may be: (i) positioned adjacent to both the rear edge **322** of the frontal shell **228** and the frontal edge **253** of the rear shell **234**, and (ii) illuminated by an extent of the illumination assembly **263**, namely by light that is emitted by at least one of the light emitters **264**.**2***a*-**264**.**2***d*. An extent of the head portion **202***a* is provided by these peripheral protrusions **262***a*-**262***d*, namely the voids **262**.**4***a*-**262**.**4***d*. The voids **262**.**4***a*-**262**.**4***d* are recessed relative to both: (i) a first location on the outer surface So of the frontal shell **228** that is positioned adjacent to the gap **370**, and (ii) a second location on the outer surface of the rear shell **234** that is also positioned adjacent to said gap **370**. The rear edge **322** of the frontal shell **228** does not directly abut the forward edge **298** of the rear shell **234** at a location corresponding to these gaps **370**. As such, recess material may be positioned in these gaps **370**, and wherein said recess material may be a portion of the peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d* that form the light emitter housings **262.10***a*-**262.10***b*. In other words, an extent of the portion of the peripheral protrusions **262***a*, **262***b*, **262***c*, **262***d* that form the light emitter housings **262**.**10***a*-**262**.**10***b* may be recessed relative to the outer surfaces of both the frontal shell **228** and the rear shell **234**. This specific positional relationship may cause an extent of the head portion **202***a* (namely, parts of the protrusions **262**) to be positioned: (i) effectively within the envelope defined by the frontal shell **228** and/or the rear shell **234**, and (ii) at said location to help connect the frontal shell **228** to the rear shell **234** (via the intermediate cover **252** to which they are attached). Light emitted from the illumination assembly **263** may obscure an extent of the head portion **202***a* (specifically, the recessed surfaces of the protrusions **262**), and may specifically obscure an extent of the head portion **202***a* that has an outer surface that is recessed relative to the outer surfaces of the frontal and rear shells **228**, **234**.

(146) When viewing the head portion **202***a* from the front, as shown in FIGS. **4** and **14**, it can be seen that the light emitters **264**.**2***a*-**264**.**2***d* of assemblies **264***a*-**264***d* are spaced apart from the display **300** and the electronics support **254**. In other words, the light emitters **264**.**2***a*-**264**.**2***d* do not reside behind or overlap with the display **300** or the electronics support **254**, although in some alternative embodiments, such an overlap may occur. The lower light emitting assemblies **264***b*, **264***d* are positioned below the display **300** and the electronics support **254**. The upper light emitting assemblies **264***a*, **264***c* flank the display **300** and the electronics support **254** such that a horizontal plane (i.e., the plane **122**) extending through these upper light emitting assemblies **264***a*, **264***c* also passes through the display **300**. This plane **122** also passes through the center C of the head portion **202***a*, but the upper light emitting assemblies **264***a*, **264***c* are depicted as being slightly offset upward relative to this plane **122**. The upper light emitting assemblies **264***a*, **264***c* are also located below a top end of the display **300** so as to be positioned below any cameras or sensors mounted to the electronics support **254** above the display **300**.

(147) iii. Sensor Assembly

(148) The sensor assembly **301***a* associated with the head portion **202***a* may include one or more cameras (such as upper cameras **302**, **303**, **304**), and/or other types of sensors including temperature sensors, pressure sensors, force sensors, inductive sensors, capacitive sensors, any combination of these sensors, or other known types of sensors. In the illustrative example provided, the sensor assembly **301***a* includes a set of upper cameras **301***a*.**2**. For example, the set of upper camera **302.1** may include three upper cameras **302**, **303**, **304** that may be positioned in the head volume, in a forward facing orientation O.sub.FF: (i) above: (a) the display/shield **300**, (b) the orbital region **368**, (c) the nasal region **357**, (d) the oral region **366**, and (e) an extent of the illumination assembly **263** and at least one light emitter **264***a*-**264***d*, (ii) generally positioned in the frontal region **362**, (iii) vertically above an extent of: (a) the temporal region **350**, (b) the zygomatic region **358**, and (c) the buccal region **352**, and (iv) directed forward. As shown in FIGS. **26** and **27**, the three upper cameras **302**, **303**, **304** are substantially horizontally aligned and spaced apart from

one another along a plane that is parallel with plane **124**. Although these upper cameras **302**, **303**, **304** are shown as illustrative examples, other types of sensors may be relied upon and coupled to the internal mounting frame (or electronics support **254**) in a similar manner to ensure proper directional positioning for their respective detection, sensing, or signal reception capabilities.

(149) The sensor assembly **301**_a_ and/or camera **302**, **303**, **304** may include: (i) scan camera(s), (ii) monochrome camera(s), (iii) color camera(s), (iv) CMOS camera(s), (v) CCD sensor(s) or camera(s) that include CCD sensor(s), (vi) camera(s) or sensor(s) that feature either a rolling shutter or a global shutter mechanism, (vii) other types of 2D digital camera(s), (viii) other types of 3D digital camera(s), (ix) camera(s) or sensor(s) that are capable of stereo vision, structured light projection, and/or laser triangulation techniques, (x) sonar camera(s) or ultrasonic camera(s), (xi) infrared sensor(s) and/or infrared camera(s), (xii) radar sensor(s), (xiii) LiDAR, (xiv) other structured light sensors, camera(s), or technologies, (xv) dot projecting camera(s) or sensor(s), or (xvi) any combination of the above-listed sensors or any other known camera or sensor type. For example, the camera **303**, **302**, **304** used in the sensor assembly **301**_a_ may have a megapixel resolution of between approximately 0.4 Megapixels (MP) to 20 MP, may be capable of recording video at frame rates from 5.6 Frames Per Second (FPS) to 286 FPS, may utilize a CMOS sensor, its pixel size may range from 2.4 micrometers (um) to 6.9 um, it may utilize a Starvis rolling shutter technology, it can operate in ambient air temperatures up to 55 degrees Celsius, and may possess any other properties, technologies, or features that are discussed within U.S. Pat. Nos. 11,402,726, 11,599,009, 11,333,954, or 11,600,010, all of which are incorporated herein by reference. It should be understood that the cameras are typically configured as video cameras but may, in alternative configurations, function primarily as still image cameras.

D. Neck Portion

(150) As shown in at least FIG. **3**, the neck portion **202**_b_ of the head and neck assembly **202** includes a deformable cover and member **230** that is designed to extend from an upper portion of the torso **204** to a lower portion of the head portion **202**_a_. In particular, this deformable cover and member **230** is configured to wrap around at least an edge portion of the rear shell **234** of the head portion **202**_a_. In doing so, the deformable cover and member **230** effectively obscures the actuators and other electronics that are contained within the neck portion **202**_b_.

E. Distances, Angles, Radii, and Arcs

(151) TABLE-US-00001 TABLE 1 Distance Lower Upper Preferred Lower Preferred Upper (mm) Bound Bound Bound Bound D6 81.5 122.3 91.7 112.1 D7 80.5 120.8 90.6 110.7 D8 76.1 114.2 85.6 104.7 D9 80.6 120.9 90.7 110.9 D10 34.2 51.2 38.4 47.0 D11 61.7 92.6 69.4 84.8 D22 93.3 140.0 105.0 128.3 D23 173.6 260.4 195.3 238.7 D24 166.2 249.4 187.0 228.6

(152) TABLE-US-00002 TABLE 2 Preferred Preferred Angle Lower Upper Lower Upper (Degrees) Bound Bound Bound Bound A1 5.7 8.6 6.4 7.9

(153) TABLE-US-00003 TABLE 3 Preferred Preferred Radius Lower Upper Lower Upper (mm) Bound Bound Bound Bound R1 99.7 149.6 112.2 137.1 R2 46.0 69.0 51.7 63.2 R10 127.7 191.6 143.7 175.6 R11 79.3 118.9 89.2 109.0 R12 69.0 103.5 77.6 94.8

(154) TABLE-US-00004 TABLE 4 Preferred Preferred Arc Lower Upper Lower Upper (Degrees) Bound Bound Bound Bound Arc2 115.1 172.6 129.5 158.2 Arc4 110.8 166.2 124.6 152.3 Arc5 117.1 175.6 131.7 161.0 Arc6 65.8 98.7 74.1 90.5 Arc7 39.6 59.5 44.6 54.5

F. Alternative Embodiments of the Head and Neck Assembly

(155) FIGS. **28-62** show seven alternative embodiments of the head and neck assembly **2202**, **3202**, **4202**, **5202**, **6202**, **7202** that may be used interchangeably in connection with the two different general embodiments of the humanoid robot **100** shown in FIGS. **1-2**, **10-11**, and **32**. Each of these alternative embodiments has slightly different structural features and configurations, but each embodiment maintains a similar and generally tapered overall shape. As discussed in more detail below, the features and/or components of each of these embodiments may be interchanged, added to, or removed from the features and/or components of other disclosed embodiments. For example,

a lower recess **3310** that is formed in the chin or mental region **355** of the third embodiment of the head portion (e.g., as shown in FIGS. **32-57**B, head **3202***a*) may be removed, similar to how such a lower recess was removed from the fifth, sixth, and seventh embodiments (e.g., FIGS. **58-62**). Additionally, the concept of utilizing a deformable material in the neck assembly (e.g., neck shell **3230**) and including actuators within the neck assembly to allow the head portion to move may be added to the head and neck assembly **202** of the first embodiment. It should be understood that these are only illustrative examples, and any feature and/or component from one embodiment may potentially be interchanged with, added to, or removed from any other embodiment disclosed herein.

1. Second Embodiment

(156) Similar to the head and neck assembly **202** described above in connection with FIGS. **1-27**, FIGS. **28-31**B illustrate a second embodiment of a head and neck assembly, designated as **2202**. For the sake of brevity, the detailed disclosure provided above in connection with the head and neck assembly **202** will not be fully repeated below, but it should be understood that like numerals in the different embodiments generally represent similar or like structures. For example, the disclosure regarding the display **300** applies equally to the display **2300** of the second embodiment. Further, it should be understood that the functionality and operation of the head and neck assembly **2202** are similar or identical to the features and functionality disclosed with respect to the head and neck assembly **202**. It should also be understood that any one or more features of the head and neck assembly **202** may be used in combination with those disclosed with respect to the head and neck assembly **2202**, and, conversely, that any one or more features of the head and neck assembly **2202** may be used in combination with those disclosed with respect to the head and neck assembly **202**. The primary difference between the head and neck assembly **202** and the head and neck assembly **2202** is the fact the housing assembly **2220** of the second embodiment further includes front and rear recessed areas **2272**, **2277** respectively, and an upper shell **2270**. These differences also contribute to a different overall shape for the frontal shell **2228** of this second embodiment.

(157) As shown in FIGS. **28-31**B, the frontal shell **2228** is configured to cover a front portion of an electronics assembly **2222**, thereby locating at least a portion of the electronics assembly **2222** between an intermediate cover **2252** and the frontal shell **2228**. In the illustrative second embodiment, the housing assembly **2220** further includes an upper shell **2270** and a sensor recess **2272**. The frontal shell **2228** may be formed to include a central notch **2274** that conforms to the shape of this sensor recess **2272**. The housing assembly **2220** is configured to hold the display **2300** and at least one sensor, such as upper cameras **2302**, which are part of the electronics assembly **2222** mounted on an electronics support **2254**. The sensor recess **2272** may be positioned above an extent of the frontal shell **2228** and is integrated within the upper shell **2270**. The housing assembly **2220** is shaped with a curved surface in at least the upper shell **2270** to resemble a head portion and has a rear facing edge **2276** that extends downward from the upper shell **2270** at a rear end thereof. For example, when viewed from the side, the housing assembly **2220** may have a rearwardly sloping, substantially linear edge with a forward angle (e.g., extending rearward from a horizontal reference) of between 90 degrees and 140 degrees, and preferably around 110 degrees from horizontal, when the humanoid robot **100** is in a normal vertical standing position (meaning the sagittal and coronal planes of the head are aligned with the sagittal and coronal planes of the robot). The upper shell **2270** may include a curved portion of the housing assembly **2220** from the rear edge to the sensor recess **2272** and may be defined by a curvilinear border **2278** that surrounds the upper shell **2270** and sensor recess **2272**. In some examples, the upper shell **2270** has a continuous surface from the rear edge to the sensor recess **2272**.

(158) The upper sensor recess **2272** includes one or more sensor openings **2304** that are set back or recessed from the front of the housing assembly **2220**. These sensor openings **2304** are positioned to correspond with the upper camera(s) **2302** of the electronics assembly **2222**. The sensor openings **2304** are partially protected by an overhang feature of the upper shell **2270** that protrudes

over these sensor openings **2304**. A shelf **2284** extends forward and downward from the sensor opening **2304** at an angle such that this shelf **2284** will not obscure the line of sight of the upper cameras **2302** (or other sensors) of the electronics assembly **2222**. The sensor recess **2272** may have a contoured surface to provide a smooth transition from the shelf **2284** to the overhang of the upper shell **2270**. The curvilinear border **2278** extends from the edge of the shelf **2284** up to the upper shell **2270**. The frontal shell **2228** of this embodiment may have a main body portion and two wing-like projections **2228.4**, **2228.6** that extend upward from this main body **2228.2**. Wherein said wing-like projections **2228.4**, **2228.6** are designed to flank the upper sensor recess **2272** and are configured to have extents that are positioned adjacent to the curvilinear border **2278** of the upper sensor recess **2272**.

(159) The housing assembly **2220** may also include a sensor cover **2286** made of a material that does not obscure a signal intended to be detected by the sensor(s) housed therein. For example, the sensor cover **2286** may be a planar cover made of a transparent material that allows the upper cameras **2302** to receive images, preferably undistorted images. As such, housing assembly **2220** further comprises an upper substantially flat region **2220.2** that is provided by the sensor cover **2286**, and wherein said upper substantially flat region **2220.2** located above the display **2300** that displays one or more visual representations (as shown in FIGS. **9**A-**9**G). Additionally and/or alternatively, the sensor cover **2286** may have openings formed directly therein for receiving an extent of a sensor (e.g., a camera lens protruding through). In further embodiments, the upper sensor recess **2272** may be omitted entirely, and in such a case, the frontal shell **2228** itself may include openings formed therein for receiving an extent of a sensor (e.g., a camera lens).

(160) In the illustrative embodiment shown in FIGS. **28-31**B, a rear sensor recess **2277** may be formed in the head portion **2202***a*, positioned between a rear end of the upper shell **2270** and the rear shell **2234**. This rear sensor recess **2277** can form a viewport for a rear facing sensor **2301***c* (e.g., camera) or can function as a vent to allow airflow into and/or out of the head portion **2202***a* for the purpose of cooling the electronics assembly **2222**, potentially via an internal fan or a plurality of fans. As shown in FIGS. **28-31**B, the frontal shell **2228** may include a curved upper surface defining the slot or notch **2274** and may be configured to fit between the curvilinear border **2278** and a rim. The frontal shell **2228** is shaped to generally resemble the form of the head portion **2202***a*, thereby providing a substantially continuous surface from the upper shell **2270** and sensor recess **2272** to the rear shell **2234**. The curvature of the frontal shell **2228** may vary and may have different curvatures (i.e., different radii and arcs) at different positions along the frontal shell **2228**. The frontal shell **2228** may include light recesses **2296***a*, **2296***b*, **2296***c*, **2296***d* to conform with the shape of the peripheral protrusions **2262***a*, **2262***b*, **2262***c*, **2262***d*.

2. Third Embodiment

(161) Similar to the head and neck assemblies **202** and **2202** described above in connection with FIGS. **1-27** and FIGS. **28-31**B respectively, FIGS. **32-57**B illustrate a third embodiment of a head and neck assembly, designated as **3202**. For the sake of brevity, the detailed disclosure provided above in connection with the head and neck assemblies **202** and **2202** will not be fully repeated below, but it should be understood that like numerals in the various embodiments generally represent similar or like structures. For example, the disclosure regarding the display **300** applies equally to the display **3300** of this third embodiment. Further, it should be understood that the functionality and operation of the head and neck assembly **3202** are similar or identical to the features and functionality disclosed with respect to the head and neck assemblies **202** and **2202**. It should also be understood that any one or more features of the head and neck assemblies **202**, **2202** may be used in combination with those disclosed with respect to the head and neck assembly **3202**, and, conversely, that any one or more features of the head and neck assembly **3202** may be used in combination with those disclosed with respect to the head and neck assemblies **202**, **2202**. The primary differences between the head and neck assembly **2202** (second embodiment) and the head and neck assembly **3202** (third embodiment) are the fact that the head portion **3202***a* of the third

embodiment does not include the rear sensor recess **2277** (found in the second embodiment), but it does include a lower sensor cover **3310** in the chin region, and it incorporates a deformable neck shell **3230**.

(162) i. Upper Shell

(163) The head portion **3202***a* of the third embodiment includes an upper shell **3270** which has a recessed sensor region **3272** at a front end of this upper shell **3270**, operating similarly to that of the housing assembly **2220** of the second embodiment. However, a rear end of the upper shell **3270** in this third embodiment is substantially flush with the rear shell **3234**, such that there is no recessed region in this particular area (unlike the rear sensor recess **2277** of the second embodiment). In some examples of this third embodiment, the upper shell **3270** may further include an additional recessed area **3280**. This recessed area **3280** may be configured to hold a top shell or cover plate **3282**, which is considered part of the upper shell **3270** and has a shape that conforms to the shape of the recessed area **3280**.

(164) ii. Neck Shell

(165) The neck shell **3230** of the third embodiment may be made from a material (e.g., fabric or a deformable plastic) that allows the head portion **3202***a* to twist in both directions (yaw) and pitch forward and back without the material bunching excessively or pulling unduly. The neck shell **3230** may be elastic or resilient in nature and is designed to return to its original, undeformed state when the head portion **3202***a* returns to its normal, neutral state or position. The disclosed head and neck assembly **3202** may include one or more actuators that allow the head portion **3202***a* to: (i) twist or rotate, and (ii) tilt or change its pitch. Unlike many conventional robots, these actuators are hidden underneath the deformable neck shield **3230**. Movement of these actuators causes the deformable neck shell **3230** to deform and thereby accommodate such movements of the head portion **3202***a*.

(166) The deformable neck shield **3230** is designed to extend up to approximately the jaw line of the head enclosure (housing assembly) and also into a rear extent of the head enclosure, but it generally does not extend into the side regions of the head portion **3202***a*. This specific configuration ensures that the neck shield **3230** is sufficiently attached to the head portion **3202***a*, yet it minimizes the surface area of the head portion **3202***a* that is covered by this deformable neck shield **3230**. Minimizing the coverage of the deformable neck shield **3230** in the side regions of the head portion **3202***a* allows for the inclusion of more durable materials in these side regions without necessitating the use of overlapping layers of materials. This design is beneficial over conventional robot heads because it can reduce overall material usage and/or increase the lateral protection afforded to the electronics contained within the head portion **3202***a*.

(167) iii. Lower Sensor Cover

(168) As shown in FIGS. **33**, **34**, and **56**, the head portion **3202***a* of the third embodiment further includes a lower recess formed in a chin region of this head portion **3202***a*, which may include a lower sensor cover **3310**. The chin region itself projects outward away from the neck portion, thereby providing a lower surface **3311** that generally faces toward the ground when the robot is upright. The lower sensor cover **3310** is coupled to this lower surface **3311** and consequently also faces generally toward the ground. The lower sensor cover **3310** is shaped to couple with the head portion **3202***a* in a position such that a sensor opening **3312**, which is included in the lower sensor cover **3310**, corresponds with a sensor **3301***b*, such as one or more lower camera(s) **3305** that are part of the electronics assembly **3222**. The lower camera **3305** is configured to monitor the ground for obstacles and/or to monitor the movement of various parts of the humanoid robot **100**. The lower sensor cover **3310** may be made of a material that does not obscure a signal intended to be detected by the sensor housed behind it. For example, the lower sensor cover **3310** may be a planar cover made of a transparent material that allows the lower cameras **3305** to receive images, preferably undistorted images. The lower sensor cover **3310** may be enclosed by the frontal shell **3228** and the neck shell **3230**.

(169) iv. Electronics Support Frame

(170) The frontal shell **3228** is shown removed from the head portion **3202***a* in FIGS. **54-58**. Both The frontal shell **3228** and the display **3300** are mounted to an electronics support frame, also referred to as a shielded portion **3288**; this electronics support frame **3288** is essentially a combination of the intermediate cover and the electronics support of the first embodiment, now integrated into a single component. The shielded portion **3288** may include a substantial surface that extends from a curvilinear border **3290** (which surrounds an extent of the upper shell **3270** and the sensor recess **3272**) down to a rim **3292**. This shielded portion **3288** may include a display opening **3294** that is positioned and sized to receive the display **3300** when mounted. The display **3300** may be rectangular in overall shape but possess a curvature. The shielded portion **3288** may be shaped with contours around this display opening **3294** to properly receive the curved shape of the display **3300** without obstructing the view of the active display area. The shielded portion **3288** may also have a taper and/or include additional contours in the region between the display opening **3294** and the rim **3292**. The rim **3292** may include lighting recesses or peripheral protrusions **3262***a*, **3262***b*, **3262***c*, **3262***d*, which are formed within this rim **3292** to receive the light emitting assemblies **3264***a*, **3264***b*, **3264***c*, **3264***d* of the electronics assembly **3222**. Although the illustrative embodiment shown in FIGS. **54-58** depicts the frontal shell **3228** as being sized to fit within the shielded portion **3288**, it should be understood that the frontal shell **3228** may occupy any portion or ratio of the robot's head portion **3202***a*, may have any suitable configuration, or in some cases, may even be omitted. In some embodiments, therefore, the frontal shell **3228** is an optional component.

3. Fourth Embodiment

(171) Similar to the head and neck assemblies **202**, **2202**, and **3202** described above in connection with FIGS. **1-27**, FIGS. **28-31**B, and FIGS. **32-57**B respectively, FIGS. **58-59** illustrate a fourth embodiment of a head and neck assembly, designated as **4202**. For the sake of brevity, the detailed disclosure provided above in connection with the head and neck assemblies **202**, **2202**, and **3202** will not be fully repeated below, but it should be understood that like numerals generally represent generally similar or like structures in the various embodiments. For example, the disclosure regarding the display **300** applies equally to the display **4300** of this fourth embodiment. Further, it should be understood that the functionality and operation of the head and neck assembly **4202** are similar or identical to the features and functionality disclosed with respect to the head and neck assemblies **202**, **2202**, and **3202**. It should also be understood that any one or more features of the head and neck assemblies **202**, **2202**, **3202** may be used in combination with those disclosed with respect to the head and neck assembly **4202**, and, conversely, that any one or more features of the head and neck assembly **4202** may be used in combination with those disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**.

(172) The primary difference between the head and neck assembly **3202** (third embodiment) and the head and neck assembly **4202** (fourth embodiment) is the fact that the shielded portion **4288** of the fourth embodiment is ribbed, and the head portion **4202***a* of this fourth embodiment lacks a lower recessed sensor region in the chin area (which was present as **3310** in the third embodiment). As shown in FIGS. **58** and **59**, the shielded portion **4288** can include a plurality of ribs **4372**, **4374** that define one or more recessed areas **4370** between them. These ribs **4372**, **4374** can provide additional structural reinforcement to areas adjacent to the electronics assembly **4222**. The recessed areas **4370** can simply be part of the structural shielded portion **4288** or, alternatively, can themselves house additional displays and/or lights that are used to convey information and indications to users. Although not explicitly shown in FIGS. **58** and **59**, the head portion **4202***a* of this fourth embodiment can either include or omit a frontal shell.

4. Fifth Embodiment

(173) Similar to the head and neck assemblies **202**, **2202**, **3202**, and **4202** described above in connection with FIGS. **1-27**, FIGS. **28-31**B, FIGS. **32-57**B, and FIGS. **58-59** respectively, FIG. **60** illustrates a fifth embodiment of a head and neck assembly, designated as **5202**. For the sake of

brevity, the detailed disclosure provided above in connection with the head and neck assemblies **202**, **2202**, **3202**, and **4202** will not be fully repeated below, but it should be understood that like numerals generally represent generally similar or like structures in the various embodiments. For example, the disclosure regarding the display **300** applies equally to the display **5300** of this fifth embodiment. Further, it should be understood that the functionality and operation of the head and neck assembly **5202** are similar or identical to the features and functionality disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**, and **4202**. It should also be understood that any one or more features of the head and neck assemblies **202**, **2202**, **3202**, **4202** may be used in combination with those disclosed with respect to the head and neck assembly **5202**, and, conversely, that any one or more features of the head and neck assembly **5202** may be used in combination with those disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**, **4202**.

(174) The primary difference between the head and neck assembly **3202** and the head and neck assembly **5202** (fifth embodiment) is the fact that the head portion **5202***a* of the fifth embodiment includes only one continuous light emitting assembly **5264**, and this head portion **5202***a* also lacks a lower recessed sensor region in the chin area. As shown in FIG. **60**, the light emitting assembly **5264** in this fifth embodiment is in the form of a band and extends along the interface **5320** between the frontal shell **5228** and the rear shell **5234**. The light emitting assembly **5264** extends along the entire interface **5320** in the illustrative embodiment shown, but it may extend only partway along this interface **5320** in some alternative embodiments. In other words, the light emitting assembly **5264** may extend completely around the periphery of the frontal shell **5228**. The outer surface of this light emitting assembly **5264** may be flush with the outer surfaces of the frontal shell **5228** and rear shell **5234**, or it may be recessed relative to one or both of these front and/or rear shells **5228**, **5234**.

5. Sixth Embodiment

(175) Similar to the head and neck assemblies **202**, **2202**, **3202**, **4202**, and **5202** described above in connection with FIGS. **1-27**, FIGS. **28-31**B, FIGS. **32-57**B, FIGS. **58-59**, and FIG. **60** respectively, FIG. **61** illustrates a sixth embodiment of a head and neck assembly, designated as **6202**. For the sake of brevity, the detailed disclosure provided above in connection with the head and neck assemblies **202**, **2202**, **3202**, **4202**, and **5202** will not be fully repeated below, but it should be understood that like numerals generally represent generally similar or like structures in the various embodiments. For example, the disclosure regarding the display **300** applies equally to the display **6300** of this sixth embodiment. Further, it should be understood that the functionality and operation of the head and neck assembly **6202** are similar or identical to the features and functionality disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**, **4202**, and **5202**. It should also be understood that any one or more features of the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202** may be used in combination with those disclosed with respect to the head and neck assembly **6202**, and, conversely, that any one or more features of the head and neck assembly **6202** may be used in combination with those disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202**. The primary difference noted for this sixth embodiment (head and neck assembly **6202**) compared to some earlier embodiments is the fact that the head portion **6202***a* of the sixth embodiment lacks a front recessed sensor region. Like the fifth embodiment, the head portion **6202***a* of the sixth embodiment includes only one continuous light emitting assembly **6264** that extends along the interface **6320** between the frontal shell **6228** and the rear shell **6234**.

6. Seventh Embodiment

(176) Similar to the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202**, and **6202** described above in connection with FIGS. **1-27**, FIGS. **28-31**B, FIGS. **32-57**B, FIGS. **58-59**, FIG. **60**, and FIG. **61** respectively, FIG. **62** illustrates a seventh embodiment of a head and neck assembly, designated as **7202**. For the sake of brevity, the detailed disclosure provided above in connection with the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202**, and **6202** will not be fully

repeated below, but it should be understood that like numerals generally represent generally similar or like structures in the various embodiments. For example, the disclosure regarding the display **300** applies equally to the display **7300** of this seventh embodiment. Further, it should be understood that the functionality and operation of the head and neck assembly **7202** are similar or identical to the features and functionality disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202**, and **6202**. It should also be understood that any one or more features of the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202**, **6202** may be used in combination with those disclosed with respect to the head and neck assembly **7202**, and, conversely, that any one or more features of the head and neck assembly **7202** may be used in combination with those disclosed with respect to the head and neck assemblies **202**, **2202**, **3202**, **4202**, **5202**, **6202**. The primary difference between the head and neck assembly **6202** (sixth embodiment) and the head and neck assembly **7202** (seventh embodiment) is the fact that the frontal shell **7228** of the seventh embodiment provides substantially all of the exterior surface of the head portion **7202***a*. As shown in FIG. **62**, the head portion **7202***a* of this seventh embodiment further includes a light emitting assembly **7264***a* that is coupled to this prominent frontal shell **7228**. The neck shell **7230** (associated with neck portion **7202***b*) can also include a second light emitting assembly **7264***b* located adjacent to an edge of the frontal shell **7228**.

G. Body Illumination Assembly

(177) As previously described in detail, the head portion **202***a* (and its variants) includes a head illumination assembly, exemplified by light emitting assembly **264***a*-**264***d* (and its variants **2264**, **3264**, etc.) located at the shield interface **320** (and its variants). The humanoid robot **100** can further include additional illumination assemblies located in other areas of the robot **100**, such as a torso illumination assembly (with emitters **330**), a thigh illumination assembly (with emitters **332**), a neck illumination assembly (with emitters **334**), a shoulder illumination assembly (with emitters **336**), a hand/wrist illumination assembly (with emitters **338**), a knee illumination assembly (with emitters **340**), and a hip illumination assembly (with emitters **342**). The illumination assemblies disclosed herein are intended to meet the standards described in FDA CFR Title 21 part 1040.10, titled “Performance standards for Light-Emitting Products,” and ANSI LIA Z136.1, titled “Safe Use of Lasers,” at the time of filing this application and are fully incorporated herein by reference.

(178) 1. Torso Illumination Assembly

(179) The torso illumination assembly includes at least one, and preferably a plurality of, light emitters **330** that are located along a front surface of the torso **204**, generally corresponding to a chest region of the humanoid robot **100**, as shown in FIG. **63**A. These torso light emitters **330** are illustratively embodied as elongated light strips; however, the light emitters **330** can have any suitable shape or structure as dictated by design requirements. The torso light emitters **330** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The torso light emitters **330** can utilize one or more of the following light technologies: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or any combination of the above, or any other known technology or light-emitting feature.

(180) 2. Thigh Illumination Assembly

(181) The thigh illumination assembly includes at least one, and preferably a plurality of, light emitters **332** that are located along a front surface of each thigh **404***a*, **404***b*, generally corresponding to a quadricep region of the humanoid robot **100**, as shown in FIGS. **63**B and **63**C. These thigh light emitters **332** are illustratively embodied as elongated light strips; however, the light emitters **332** can have any suitable shape or structure. The thigh light emitters **332** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The thigh light emitters **332** can utilize one or more of the following light technologies: LCD, LED,

OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or any combination of the above, or any other known technology or light-emitting feature.

(182) 3. Neck Illumination Assembly

(183) The neck illumination assembly includes at least one, and preferably a plurality of, light emitters **334** that are located along a rear surface of the neck portion **202***b*, specifically at the base of the head portion **202***a*, as shown in FIG. **63**A. This neck light emitter (or emitters) **334** is illustratively embodied as an elongated light strip; however, the light emitters **334** can have any suitable shape or structure. The neck light emitters **334** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The neck light emitters **334** can utilize one or more of the following light technologies: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or any combination of the above, or any other known technology or light-emitting feature.

(184) 4. Shoulder Illumination Assembly

(185) The shoulder illumination assembly includes at least one, and preferably a plurality of, light emitters **336** that are located along a front surface of each shoulder **206***a*, **206***b*, as shown in FIG. **65**A. These shoulder light emitters **336** are illustratively embodied as elongated light strips; however, the light emitters **336** can have any suitable shape or structure. The shoulder light emitters **336** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The shoulder light emitters **336** can utilize one or more of the following light technologies: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or any combination of the above, or any other known technology or light-emitting feature.

(186) 5. Hip Illumination Assembly

(187) The hip illumination assembly includes at least one, and preferably a plurality of, light emitters **342** that are located along a surface of each hip, as shown in FIG. **65**B. These hip light emitters **342** are illustratively embodied as elongated light strips; however, the light emitters **342** can have any suitable shape or structure. The hip light emitters **342** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The hip light emitters **342** can utilize one or more of the following light technologies: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or any combination of the above, or any other known technology or light-emitting feature.

(188) 6. Hand Illumination Assembly

(189) The hand illumination assembly includes at least one, and preferably a plurality of, light emitters **338** that are located along a surface of each hand **216***a*, **216***b*, as shown in FIG. **65**B. These hand light emitters **338** are illustratively embodied as elongated light strips; however, the light emitters **338** can have any suitable shape or structure. The hand light emitters **338** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The hand light emitters **338** can utilize one or more of the following light technologies: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or

any combination of the above, or any other known technology or light-emitting feature.

(190) 7. Knee Illumination Assembly

(191) The knee illumination assembly includes at least one, and preferably a plurality of, light emitters **340** that are located along a surface of each knee **406***a*, **406***b*, as shown in FIG. **65**C. These knee light emitters **340** are illustratively embodied as elongated light strips; however, the light emitters **340** can have any suitable shape or structure. The knee light emitters **340** can be operated in the same way as the head light emitting assembly **264***a*-**264***d* described above. The knee light emitters **340** can utilize one or more of the following light technologies: LCD, LED, OLED, LPD, IMOD, QDLED, mLED, AMOLED, SED, FED, plasma, electronic paper or EPD, MicroLED, quantum dot display, LED backlit LCD, WLCD, OLCD, transparent OLED, PMOLED, capacitive touchdisplay elements, resistive touchdisplay elements, monochrome emitters, color emitters, or any combination of the above, or any other known technology or light-emitting feature.

H. Other Electronic Components

(192) As described above, the electronic components of the head may also include a directional microphone, speaker, antennas, light emitting assembly **264***a*-**264***d*, as well as a data storage device and/or computing device comprising a processor and memory. Specifically, the directional microphone is designed to detect sounds and determine a position, which enables the robot to move its head toward the sound. In particular, one or more speakers may be configured to allow the robot to communicate with nearby humans with audible messages or responses. One or more antennas may be configured to transmit and receive data wirelessly for data transfer into and out of the robot. Specifically, the robot may include wireless communication modules (e.g., cellular, Wi-Fi, Bluetooth, WiMAX, HomeRF, Z-Wave, Zigbee, THREAD, RFID, NFC, and/or etc.) that are connected to the antennas. For example, the robot head portion **202***a* may include a 5G cellular radio coupled to one of the antennas and a Wi-Fi radio (e.g., 5 GHz or 2.4 GHz) coupled to the other antenna.

(193) The data storage device may include a solid-state hard drive designed to capture all of the data generated by the sensors or a subset of the data generated by the sensors. The subset of the data may be time-based (e.g., the pre-defined time surrounding the start up/shut down of the robot), sensor-based (e.g., only encoder data), movement/configuration-based (e.g., when performing a specific task that requires the robot to put its body in a particular position/configuration), environment-based (e.g., when the robot recognizes a specific item or issue in its environment), or configuration based, error based, or a combination thereof. In addition, the data storage device may be used to store data to train other robots or store data for diagnostic purposes or any other purpose. Finally, the indicator lights may be designed to work with the display **300** to indicate a state of the robot **100** (e.g., working, error, moving, etc.) to a nearby human or may illuminate for other reasons.

(194) It is to be understood that the disclosure is not limited to the exact details of construction, operation, exact materials or embodiments shown and described, as obvious modifications and equivalents will be apparent to one skilled in the art. While the specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the disclosure, and the scope of protection is only limited by the scope of the accompanying Claims. It should also be understood that substantially utilized herein means a deviation that is less than 15% and preferably less than 5%. It should also be understood that other configuration or arrangements of the above described components is contemplated by this Application.

(195) In this patent, to the extent any U.S. patents, U.S. patent applications, or other materials (e.g., articles) have been incorporated by reference, the text of such materials is only incorporated by reference to the extent that no conflict exists between such material and the statements and drawings set forth herein. In the event of such conflict, the text of the present document governs, and terms in this document should not be given a narrower reading in virtue of the way in which

those terms are used in other materials incorporated by reference.

(196) It is to be understood that the disclosure is not limited to the exact details of construction, operation, exact materials, or embodiments shown and described, as obvious modifications and equivalents will be apparent to one skilled in the art; for example, the photographs may be digital photographs or paper based photographs that may then be scanned into digital form. While the specific embodiments have been illustrated and described, numerous modifications come to mind without significantly departing from the spirit of the disclosure.

(197) As used herein, the terms “component,” “system” and the like in relation to discussions about computer-related processes and systems are intended to refer to a computer-related entity, either hardware, a combination of hardware and software, software, or software in execution. For example, a component may be, but is not limited to being, a process running on a processor, a processor, an object, an instance, an executable, a thread of execution, a program, a computer, or both. By way of illustration, both an application running on a computer and the computer can be a component. One or more components may reside within a process, a thread of execution, or both, and a component may be localized on one computer and/or distributed between two or more computers.

(198) Furthermore, all or portions of the computer-related processes and systems can be implemented as a method, apparatus or article of manufacture using standard programming and/or engineering techniques to produce software, firmware, hardware, or any combination thereof to control a computer to implement the disclosed innovation. The term “article of manufacture” as used herein is intended to encompass a computer program accessible from any computer-readable device or media. For example, computer readable media can include but are not limited to magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips . . .), optical disks (e.g., compact disk (CD), digital versatile disk (DVD) . . .), smart cards, and flash memory devices (e.g., card, stick, key drive . . .). Additionally, it should be appreciated that a carrier wave can be employed to carry computer-readable electronic data such as those used in transmitting and receiving electronic mail or in accessing a network such as the Internet or a local area network (LAN). Of course, those skilled in the art will recognize many modifications may be made to this configuration without departing from the scope or spirit of the claimed subject matter.

(199) Where the above examples, embodiments and implementations reference examples, it should be understood by those of ordinary skill in the art that other helmet and manufacturing devices and examples could be intermixed or substituted with those provided. In places where the description above refers to particular embodiments of helmets and customization methods, it should be readily apparent that a number of modifications may be made without departing from the spirit thereof and that these embodiments and implementations may be applied to other to helmet customization technologies as well. Accordingly, the disclosed subject matter is intended to embrace all such alterations, modifications and variations that fall within the spirit and scope of the disclosure and the knowledge of one of ordinary skill in the art.

Claims

1. A humanoid robot comprising: an upper region including: (i) a torso, (ii) a pair of arm assemblies coupled to the torso, and (iii) a head and neck assembly coupled to the torso and having a neck portion and a head portion coupled to the neck portion, and wherein the head portion includes: a head housing assembly with a rear shell, and a frontal shell coupled to the rear shell to define a head volume between the frontal shell and the rear shell, and wherein the frontal shell includes a first edge extent and an opposed second edge extent, an electronics assembly with a display located in the head volume between the frontal shell and the rear shell, and wherein when the head housing assembly is in a forward facing orientation, the frontal shell includes: a substantially horizontal first arc length that extends from the first edge extent to the second edge extent at a first location below

the display, a substantially horizontal second arc length that extends from the first edge extent to the second edge extent at a second location aligned with a portion of the display, and wherein said second arc length is greater than the first arc length, and a substantially horizontal third arc length that extends from the first edge extent to the second edge extent at a third location above the display, and wherein said third arc length is greater than both the first arc length and the second arc length; and a lower region coupled to the upper region and spaced apart from the upper region, the lower region including a pair of legs.

2. The humanoid robot of claim 1, wherein the frontal shell further includes a substantially horizontal fourth arc length that extends from the first edge extent to the second edge extent at a fourth location positioned above the third arc length, and wherein the fourth arc length is greater than the third arc length.

3. The humanoid robot of claim 1, further comprising a first camera positioned above the display and within the head volume.

4. The humanoid robot of claim 3, further comprising a second camera positioned below the display and within the head volume.

5. The humanoid robot of claim 1, wherein the frontal shell and the display both have a curvilinear configuration that is concave relative to the rear shell.

6. The humanoid robot of claim 1, further comprising an illumination assembly configured to illuminate an extent of a region positioned adjacent to a first edge extent of the frontal shell.

7. The humanoid robot of claim 6, wherein an illumination gap is positioned between the frontal shell and the rear shell, and wherein light emitted from the illumination assembly is visible in said gap.

8. The humanoid robot of claim 1, wherein a portion of the frontal shell includes a tinted coating.

9. A humanoid robot comprising: an upper region including: (i) a torso, (ii) a pair of arm assemblies coupled to the torso, and (iii) a head and neck assembly coupled to the torso and having a neck portion and a head portion coupled to the neck portion, and wherein the head portion includes: a housing assembly with a rear shell and a frontal shell coupled to the rear shell to define at least one head volume between the frontal shell and the rear shell, and wherein the frontal shell has a lower frontal shell width that extends between opposed shell edges, an electronics assembly positioned within the head volume and including: (i) a display with a width that: (a) extends between opposed display edges, (b) is positioned above the lower frontal shell width, and (c) is greater than the frontal shell width, and (ii) at least one sensor, and wherein said display is configured to show one or more visual representations that do not include human-like facial features to denote an operating status of the humanoid robot, and wherein the one or more visual representations are visible through the frontal shell; and a lower region coupled to the upper region and including a pair of leg assemblies.

10. The humanoid robot of claim 9, further comprising an illumination assembly configured to emit light in a location that is adjacent to an extent of the periphery of the frontal shell.

11. The humanoid robot of claim 10, wherein the illumination assembly is configured to convey the operational status of the humanoid robot.

12. The humanoid robot of claim 9, wherein the housing assembly further comprises an upper substantially flat region located above said one or more visual representations.

13. The humanoid robot of claim 12, wherein the head portion further comprises a camera aligned with the upper substantially flat region, and wherein the upper substantially flat region includes a sensor opening configured to receive a lens of the camera.

14. The humanoid robot of claim 9, further comprises a lower camera positioned below the display.

15. The humanoid robot of claim 9, wherein the display is curved.

16. The humanoid robot of claim 9, wherein the neck portion includes a neck shell having a deformable material.

17. The humanoid robot of claim 9, further comprising a plurality of cameras, and wherein a first

camera contained in the plurality of cameras is substantially horizontally aligned with a second camera contained in the plurality of cameras.

18. A humanoid robot comprising: an upper region including: (i) a torso, (ii) a pair of arm assemblies coupled to the torso, and (iii) a head and neck assembly coupled to the torso and having a neck portion and a head portion coupled to the neck portion, and wherein the head portion includes: a housing assembly having a frontal shell and defining a head volume, and an electronics assembly including a curved display located in the head volume, and wherein a gap is formed between the curved display and the frontal shell and said curved display is configured to show one or more visual representations that do not include human-like facial features to denote an operating status of the humanoid robot; and a lower region coupled to the upper region and included a pair of leg assemblies; and further comprising an illumination assembly configured to emit light in a location that is adjacent to an extent of the periphery of the frontal shell; wherein an extent of the illumination assembly is positioned in a temporal region of the housing assembly.

19. The humanoid robot of claim 18, wherein the frontal shell includes: a first width defined by opposed edges of the frontal shell at a first location along a first substantially horizontal plane located below the display, a second width defined by opposed edges of the frontal shell at a second location along a second substantially horizontal plane passing through the display, and wherein said second width is greater than the first width, and a third width defined by opposed edges of the frontal shell at a third location along a third substantially horizontal plane located above the display, and wherein said third width is greater than both the first width and the second width.

20. The humanoid robot of claim 18, wherein the display has an upper end and an opposed lower end, and wherein the display has a constant height between the upper end and the lower end.

21. The humanoid robot of claim 20, wherein the display has a constant arc length between the upper end and the lower end.

22. The humanoid robot of claim 18, wherein the display occupies an ocular region, a nasal region, and an infraorbital region of the head portion.

23. The humanoid robot of claim 22, wherein the display further occupies at least a portion of: a frontal region, a temporal region, and a zygomatic region of the head portion.